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Bow-Tie Architecture in the
Drosophila melanogaster
Connectome:
A Two-Scale Structural Analysis

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Declaration

I hereby declare and confirm that this thesis is entirely the result of my own original work. Where other sources of information have been used, they have been indicated as such and properly acknowledged. I further declare that this or similar work has not been submitted for credit elsewhere.

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Abstract

Understanding how cognitive functions emerge from the physical wiring of the brain is one of the central challenges of modern neuroscience. The recent release of the synapse-resolution connectome of the adult *Drosophila melanogaster* brain has made it possible, for the first time, to address this question computationally on a complete nervous system.

This thesis asks whether the *Drosophila* connectome exhibits a bow-tie, or hourglass, architecture: whether many input pathways converge onto a small set of intermediate elements which then diverge again towards many outputs. The question is addressed at two complementary scales that serve as a check on each other. At the local scale we count, exhaustively and exactly, motifs in which k_{in} groups of neurons converge onto a single intermediate group, called the waist, which in turn projects onto k_{out} further groups. The counting uses a flow-vector formulation that never enumerates the search space, which is what makes an exhaustive treatment possible at all. At the global scale we compute the τ -core of the network, the smallest set of neurons carrying a prescribed fraction of all sensory-to-motor paths, and from it the hourglass score defined by Sabrin et al.

The thesis makes four contributions. The first is theoretical: we characterise exactly when motif counts compose, showing that the boundary between patterns countable with a linear flow vector and patterns requiring a sum-of-products formulation is precisely the presence of branching, and that cyclic patterns are not factorisable over vectors at all. The second is computational: a search engine that makes exhaustive counting tractable at connectome scale, gaining a factor of 219 on the worst bottleneck through a sparse reformulation, and verified row by row against an independent reference implementation. The third is empirical: an analysis of all four level windows of the hierarchy rather than a single one, which reveals that the architecture changes qualitatively with depth, with signatures involving a purely lateral branch growing from 38.0% to 82.6% of all motif instances. The fourth is methodological: a hypothesis carried through to the point where it can be tested rather than asserted. An apparent neurochemical difference between backward and forward waists is significant on the shallow window, is traced there to regional composition, and does not replicate across the remaining windows; it is therefore reported as an open question about the optic lobe rather than as a property of the connectome, together with the procedure that established the distinction.

The hourglass effect is present but depends on how paths are routed, with a score of 0.498 under structural routing against values between 0.74 and 0.87 reported for *C. elegans*. The two scales agree on the mid-hierarchy waists of the olfactory pathway and disagree on input groups, and that disagreement turns out to be interpretable rather than contradictory.

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Glossary

AL	Antennal Lobe, the first olfactory processing centre
BLAS	Basic Linear Algebra Subprograms
BWD	Backward, an edge running from a deeper to a shallower level
E/I	Excitatory / Inhibitory
EB	Ellipsoid Body
EM	Electron Microscopy
FB	Fan-shaped Body, a neuropil of the central complex
FDR	False Discovery Rate
FWD	Forward, an edge running from a shallower to a deeper level
GNG	Gnathal Ganglia
LAT	Lateral, an edge between neurons of the same level
LH	Lateral Horn
LO	Lobula, a visual processing neuropil
LOP	Lobula Plate
MB	Mushroom Body, the centre for olfactory learning
MB_CA	Mushroom Body Calyx, its input region
ME	Medulla, the largest visual neuropil
PLP	Posterior Lateral Protocerebrum
PRW	Prow
PVLP	Posterior Ventrolateral Protocerebrum
$\Sigma\Pi$	The sum-of-products formulation used for branching patterns

1 Introduction

1.1 Motivation

One of the most ambitious goals of modern neuroscience is to understand how cognitive functions such as memory, learning and decision-making emerge from the physical structure of the brain [1]. For most of the history of the field this question could only be approached indirectly. Lesion studies, electrophysiology and imaging observed neural activity without access to the underlying wiring diagram, which meant that structural hypotheses about brain function could be formulated but not rigorously tested.

Advances in large-scale electron microscopy and automated image segmentation have begun to dissolve this barrier. Connectomics, the science of producing complete maps of synaptic connectivity in neural tissue [3], now yields global wiring diagrams in which every neuron and every synapse is labelled. Where traditional neural sampling reconstructed a few hundred neurons at a time, a connectome provides the entire network at once, which permits computational analyses at a scale and resolution that were previously out of reach. The adult *Drosophila melanogaster* brain has emerged as the premier model organism for this pursuit: it contains only about 140,000 neurons, some five orders of magnitude fewer than the human brain, and yet supports a behavioural repertoire spanning visual pursuit, olfactory learning, sleep regulation and courtship [8].

The availability of a complete directed graph of a complex brain creates an opportunity, but it does not by itself create understanding. A raw connectome is a sparse, cyclic and heterogeneous directed graph with no intrinsic coordinate system. Artificial neural networks are designed with layered topologies through which information flows in one direction; biological circuits are not. They feature extensive recurrent connectivity, and feedback projections from deeper processing stages back to earlier ones are plentiful relative to feed-forward ones [14]. This makes it structurally difficult to say what “earlier” and “later” mean in a circuit, and consequently difficult to compare connectivity patterns across brain regions in any principled way, because there is no agreed axis along which to place them.

A central concept for making sense of this complexity is that of network motifs: small recurring connectivity patterns that occur in a network far more often than chance would predict [10]. At the level of neural circuits, motifs are the elementary computational building blocks from which larger structures are assembled. A feed-forward chain of three neurons implements a simple relay; adding a feedback arc to that chain turns it into a loop capable of sustaining activity over time; a

lateral connection between two populations firing in parallel creates a cross-stream interaction. If one knows which motifs are common in a given connectome, and how their composition changes from one brain region to another, it becomes possible to read circuit-level computational strategies off the wiring diagram alone.

This thesis focuses on one particular architectural template, the bow-tie, also called the hourglass. A bow-tie is a structure in which many inputs converge onto a narrow set of intermediate elements, called the waist, which then diverge again towards many outputs. The template is known to recur across widely different complex systems: cellular metabolism, the Internet protocol stack and manufacturing supply networks all display it [11]. Its presence is informative because it implies that the system compresses information through a narrow channel before redistributing it, and that in turn constrains both what the system can compute and how vulnerable it is to the loss of individual components. If the *Drosophila* brain is organised this way, the neurons occupying the waist are the ones through which most of the sensory-to-motor traffic must pass, which makes them natural targets for functional investigation.

1.2 Problem Statement

The technical question of this thesis can be stated as follows. Given the full synaptic graph of the adult *Drosophila* brain, comprising 134,181 neurons with an assigned hierarchical level and 2,700,513 synaptic pairs, determine whether the network exhibits a bow-tie architecture, at what scale it does so, and which anatomical structures occupy the waist. The problem decomposes into three sub-problems, each of which has to be solved before the next becomes meaningful.

The first is that of counting without enumeration. A bow-tie motif with two input groups and two output groups, searched over the groups of a single level window, induces a space of more than eight billion candidate combinations. Explicit enumeration of subgraph occurrences, in the style of the classical subgraph isomorphism algorithms [13, 4], is entirely infeasible at that scale. What is required instead is a counting formulation that produces exact occurrence counts without ever materialising an individual match. Such formulations do exist, but they do not exist for every pattern, and the conditions under which they apply have to be established rather than assumed. The condition turns out to be a property of the shape of the pattern and nothing else: a count factorises when the pattern contains no cycle, and does not when it contains one.

The second sub-problem concerns the relationship between two different ways of asking the same question. Motif counting is inherently local: it measures convergence

and divergence in the immediate neighbourhood of a group of neurons. The hourglass effect, by contrast, is a global property of end-to-end information flow through the whole network. These are two different measurements of the same architectural intuition, and there is no guarantee that they agree. Establishing whether they do, and characterising precisely where they do not, is part of the problem rather than an afterthought, because a disagreement between two independent measurements is either a warning that one of them is wrong or a finding in its own right.

The third sub-problem is that of distinguishing structure from composition. Anatomical regions of the fly brain differ systematically in size, in connection density and in neurotransmitter composition. Any apparent difference between two classes of motifs may therefore reflect merely *where* those motifs happen to live rather than *what* they are. Controlling for regional composition is essential, and so is repeating the controlled comparison on data that did not suggest the hypothesis in the first place. This thesis contains a case in which a controlled effect held on one subset of the data, reversed on another and disappeared on the whole, which is a pattern that looks convincingly significant at every stage until the last one.

1.3 Research Objectives

The first objective is to establish the algebra of pattern counting, by determining which connectivity patterns admit a linear counting formulation, which require a sum-of-products formulation, and which require quadratic memory. The result should be a statement about the shape of the pattern rather than an empirical rule of thumb, and it should be verifiable by exhaustive enumeration on small graphs so that its correctness does not rest on the derivation alone.

The second objective is to make exhaustive counting tractable, by designing and implementing a search engine capable of counting all bow-tie motifs over the full connectome, for every level window, in hours rather than days. Here correctness must be verified against an independent reference implementation, not merely inferred from the mathematics, because an optimisation that is correct in principle can still be wrong in code.

The third objective is to characterise the architecture at both scales, by applying the local motif search and the global hourglass analysis to the same connectome, reporting where the two converge and where they diverge, and interpreting the divergence rather than explaining it away.

The fourth objective is to test structural hypotheses with adequate controls. This means evaluating the statistical significance of the observed motifs against several complementary null models rather than one, and testing biological hypotheses

about the waists, in particular whether backward and forward waists differ in neurotransmitter composition, with explicit controls for regional composition and with replication across independent level windows.

1.4 Contributions

The theoretical contribution of this thesis is an exact characterisation of when motif counts compose, developed in Section 3.1.3. Counting occurrences of a pattern means counting the ways it can be laid over the graph, that is, the ways of assigning a neuron to each node of the pattern so that every edge of the pattern lands on an edge of the graph; these assignments are called *homomorphisms*. Both the cost of counting them and the way counts combine depend only on the shape of the pattern. Paths admit a pure matrix product and their counts add across unions of node groups; trees require an element-wise product at every branching node and their counts do not add; cyclic patterns cannot be factorised over vectors at all. The boundary between the tractable and the intractable is therefore exactly the presence of branching, and an exhaustive catalogue up to four nodes shows that 87% of connected patterns already fall on the intractable side.

The computational contribution is a search engine that is both fast and verified, described in Section 3.5. Reformulating the inner loop as a single matrix product delegated to optimised linear algebra routines, and then switching to a sparse product for the largest waists, yields a factor of 219 on the worst bottleneck of the search. The engine is verified row by row, across all 1,043,872 rows and all 26 columns, against a slow but readable reference implementation retained specifically for that purpose.

The empirical contribution is an analysis across all four level windows of the hierarchy, presented in Section 4.4. Previous work on this connectome has concentrated on a single window. Extending the analysis to all four shows that the structure changes qualitatively with depth: signatures involving a purely lateral branch account for 38.0% of all motif instances in the shallow window and 82.6% in the deep one, and the deepest level of the hierarchy turns out to be the most internally dense while remaining marginal as a waist.

The methodological contribution is a hypothesis pursued to the point where the data can decide it, reported in Section 4.9. The hypothesis is that backward waists carry a distinct neurochemical signature, and it is a natural one: inhibition is widely associated with feedback control. It is significant on the shallow window, reverses sign on the second and does not hold on the full connectome, and the reason is identified: the two classes of motif sit in different brain regions, whose neurochemistry differs.

What survives is a narrower and better specified claim about the medulla, stated as a hypothesis for future work, together with the three-step procedure that separated it from the broader one. The procedure is the transferable part, and it is described in enough detail to be reused.

1.5 Thesis Outline

Section 2 introduces the background required for the rest of the work: the graph formalism, the *Drosophila* connectome and its hierarchical decomposition, network motifs, and the hourglass framework of Sabrin et al. Section 3 develops the counting theory, both analysis pipelines, the statistical validation scheme and the computational optimisations that make the analysis feasible. Section 4 presents the results, and Section 5 summarises the contributions, states the limitations and outlines directions for further work.

2 Theoretical Background

This chapter assembles the four things the rest of the thesis needs, in the order in which they become necessary.

It begins with the graph formalism and the matrix representation of a network, which is what turns questions about connectivity into arithmetic that a computer can do at this scale. It then describes the connectome itself: what FlyWire measured, how its neurons are ordered along a sensory-to-motor axis, what the anatomical labels mean, and how neurons are grouped so that the search space becomes finite. It introduces network motifs, the notion that recurring small patterns are the building blocks of a network, and the classification of those patterns by direction that this thesis relies on. Finally it sets out the bow-tie and the hourglass, which are the two objects the whole analysis is about, and explains why they are not the same thing even though the literature often uses the words interchangeably.

That last distinction organises everything that follows. A *bow-tie* is a local pattern: a few groups converging on one and then diverging again. An *hourglass* is a global property: the whole network routing its traffic through a narrow set of intermediaries. A network can have many of the first without being the second, which is why this thesis measures both and then asks whether they agree.

2.1 Fundamentals of Network Science

Two things are needed before the connectome can be discussed at all: a language for describing networks, and a representation that makes computation on them possible. This section provides both, and neither is developed further than the rest of the thesis actually uses.

2.1.1 Graph Formalism

A connectome is modelled as a directed graph $G = (V, E)$, where V is the set of neurons and $E \subseteq V \times V$ the set of synaptic connections. An edge $(u, v) \in E$ indicates that neuron u forms at least one chemical synapse onto neuron v ; the direction of the edge is the direction in which the signal travels. Edges may additionally carry a weight $w(u, v)$ equal to the number of individual synaptic contacts between the two neurons, which in this dataset ranges from the reporting threshold of five up to several hundred.

Throughout this thesis we work with the unweighted, de-duplicated graph, so that an ordered pair of neurons contributes at most one edge regardless of how many synapses connect them. This choice is deliberate and worth justifying, because it discards information. Motif counting asks whether a connectivity *pattern* exists between particular neurons, not how strong that connection is. If multiplicities were retained, a motif count would conflate the number of distinct circuits realising a pattern with the strength of the individual connections within them, and a single very strongly connected pair could dominate the count of a pattern that occurs only once. Since the quantity of interest here is how many distinct circuits implement a given architecture, the unweighted graph is the correct object.

2.1.2 Matrix Representations

A graph can equally well be written as a matrix, and doing so is what makes the analysis in this thesis possible, so it is worth setting out both the representation and the one identity that is used throughout.

The *adjacency matrix* $A \in \{0, 1\}^{|V| \times |V|}$ records the edges in a table: the entry A_{uv} is 1 if the edge (u, v) is present and 0 otherwise. Written out in full this table is unusable at connectome scale. With 134,181 neurons it has more than 18 billion entries, occupying over 18 GB at one byte each, while the graph actually contains 2,700,513 edges: fewer than one entry in six thousand is non-zero. Storing only the non-zero entries, as the Compressed Sparse Row format does, is therefore not an optimisation but a precondition for holding the data in memory at all, and every

matrix in this thesis is stored that way.

The reason the representation matters, beyond storage, is that it turns a question about connectivity into a multiplication. Consider the product A^2 . Its entry $(A^2)_{uv}$ is $\sum_w A_{uw}A_{wv}$, and each term of that sum is 1 exactly when there is an edge from u to w and one from w to v , that is, when w is an intermediate step on a two-edge route from u to v . The sum therefore counts those routes. The same argument applied repeatedly gives the general form: the number of directed paths of length k from a set of neurons S to a set of neurons T is

$$\mathbf{1}_S^\top A^k \mathbf{1}_T, \tag{1}$$

where $\mathbf{1}_S$ is the indicator vector of S , the vector with a 1 in the position of every neuron in S and 0 elsewhere. Counting paths, an apparently combinatorial task, has become a matrix expression.

The practical value of this lies in how the expression is evaluated. Forming A^k explicitly would be hopeless, since even A^2 is far denser than A and A^3 denser still. But the expression never needs to be evaluated that way: one starts from the vector $\mathbf{1}_S$ and multiplies it by A repeatedly, obtaining after each multiplication a vector whose entry at neuron v is the number of paths of that length reaching v . Only one vector of $|V|$ numbers is ever held, and each multiplication touches each edge once, so the whole computation costs $O(k \cdot |E|)$ time and $O(|V|)$ memory instead of the cube of the number of neurons. An entire class of counting problems thereby becomes linear in the size of the network, and this vector, propagated step by step, is what Section 3.1 will call the *flow vector*.

Two cautions belong here. The identity counts *walks* rather than paths, since a route that visits the same neuron twice is included, and in a graph with cycles the distinction is not cosmetic: the number of walks grows without bound as their permitted length increases. And the trick works because a chain is a sequence with no branches, so it does not carry over unchanged to patterns that branch, which is the problem Chapter 3 takes up.

2.2 The *Drosophila melanogaster* Connectome

This section describes the object of study. It covers why the fruit fly is the organism for which this analysis is possible at all, what the FlyWire dataset contains, how its neurons are ordered along an axis running from sensory input to motor output, what the anatomical labels mean, and how neurons are grouped so that an exhaustive search becomes finite. The last two are the choices on which every later result

depends, and they are argued for rather than assumed.

2.2.1 A Model Organism for Structural Analysis

The adult *Drosophila* brain occupies a favourable position in the space of model organisms. It is small enough to have been reconstructed completely at synaptic resolution, which no vertebrate brain has been, and yet complex enough to support behaviour that is not trivially reflexive. With approximately 140,000 neurons it retains sensory integration, associative learning and memory, so that structural findings about it are plausibly informative about principles that generalise, rather than being peculiarities of a simple nervous system.

2.2.2 The FlyWire Dataset

The data used in this work derive from the FlyWire reconstruction of the adult female brain [8], distributed through the Codex portal. Four tables are used. The first lists synaptic pairs with their contact counts, already filtered at a threshold of five synapses in the public export, which removes the least reliable segmentation artefacts. The second gives neuron identifiers together with their anatomical group assignment and their predicted neurotransmitter, discussed below. The third provides functional *super-class* labels, which sort neurons by the role they play rather than by where they sit: sensory, optic, central, descending and five others, set out in full in Section 2.2.5. The fourth supplies the hierarchical level assignment described below.

The tables cover 139,255 neurons. After restricting the data to those that possess both a level assignment and a non-degenerate anatomical group, the working network comprises 134,181 neurons and 2,700,513 de-duplicated synaptic pairs. All figures reported in this thesis refer to that restricted network.

The neurotransmitter annotation requires a word of its own, since Section 4.9 depends on it entirely and it is the one field in the dataset that is inferred rather than observed. A neuron’s neurotransmitter cannot be read off an electron micrograph directly; FlyWire assigns it with a classifier trained on the appearance of the synapses themselves, and each assignment carries a confidence score. Six neurotransmitters occur in the working network: acetylcholine, in 82,298 neurons, glutamate in 19,605, GABA in 16,017, and the three modulators serotonin, dopamine and octopamine in a further 1,677 together.

These are grouped into three functional classes, plus a fourth for the uncertain. Acetylcholine is the principal *excitatory* transmitter of the insect brain, making the receiving neuron more likely to fire. GABA and glutamate are treated as *inhibitory*, making it less likely; the second of these is worth flagging, because glutamate is

excitatory in vertebrates and it is specifically in insects that it acts through a chloride channel and inhibits. Serotonin, dopamine and octopamine are *modulatory*: rather than exciting or inhibiting directly they change how other connections behave. Finally, any neuron whose prediction carries a confidence below 0.5 is treated as *unknown* and excluded from the excitatory and inhibitory counts rather than guessed at. Applying that threshold leaves 82.2% of the working network classified: 57.0% excitatory, 24.3% inhibitory, 0.9% modulatory and 17.8% unknown. That last figure is not negligible, and Section 4.9 states where it bears on the conclusions.

2.2.3 Hierarchical Levels

The raw connectome has no intrinsic notion of depth, and without one it is impossible to say whether a given connection runs forwards or backwards. To provide such a notion, each neuron is assigned a discrete hierarchical level $y \in \{1, \dots, 5\}$ reflecting its position along the axis running from sensory input to motor output. Level 1 contains the neurons closest to the sensory periphery and level 5 those closest to motor output. The level assignment used here is taken as given from earlier work on this dataset and is not re-derived. It is, however, examined from two directions: Section 4.1 reports what the levels turn out to contain, and Section 4.2 asks whether they remain informative under the grouping this thesis actually uses, which is not the one they were built for.

Because everything that follows is conditioned on these levels, it is worth stating how they were obtained. The assignment proceeds in two steps.

A spectral embedding of the connectome first places every neuron on a continuous axis. The method is called spectral because the positions are obtained from the eigenvectors of a matrix built from the connectivity, rather than from any property of the individual neurons; concretely, the construction minimises an energy that penalises edges running backwards along the axis, so that a neuron receiving from few others and projecting to many is pushed towards one end and a neuron in the opposite situation towards the other. The output is a real number for each neuron, which we will call its *flow score*, and not yet a level: it is a position on a line, with no natural place to draw a boundary.

The second step draws those boundaries. A decision tree is fitted to the flow score with the functional super-class as the target, and the thresholds it selects become the level boundaries. Because the tree is choosing the cuts that best separate sensory from optic from central from motor neurons, the boundaries are placed where the biological composition of the population changes rather than at arbitrary quantiles. Four cuts produce five levels, at flow scores 29.8, -1.5 , -13.5 and -25.4 , and Figure 1

shows both the tree and where its cuts fall on the population.

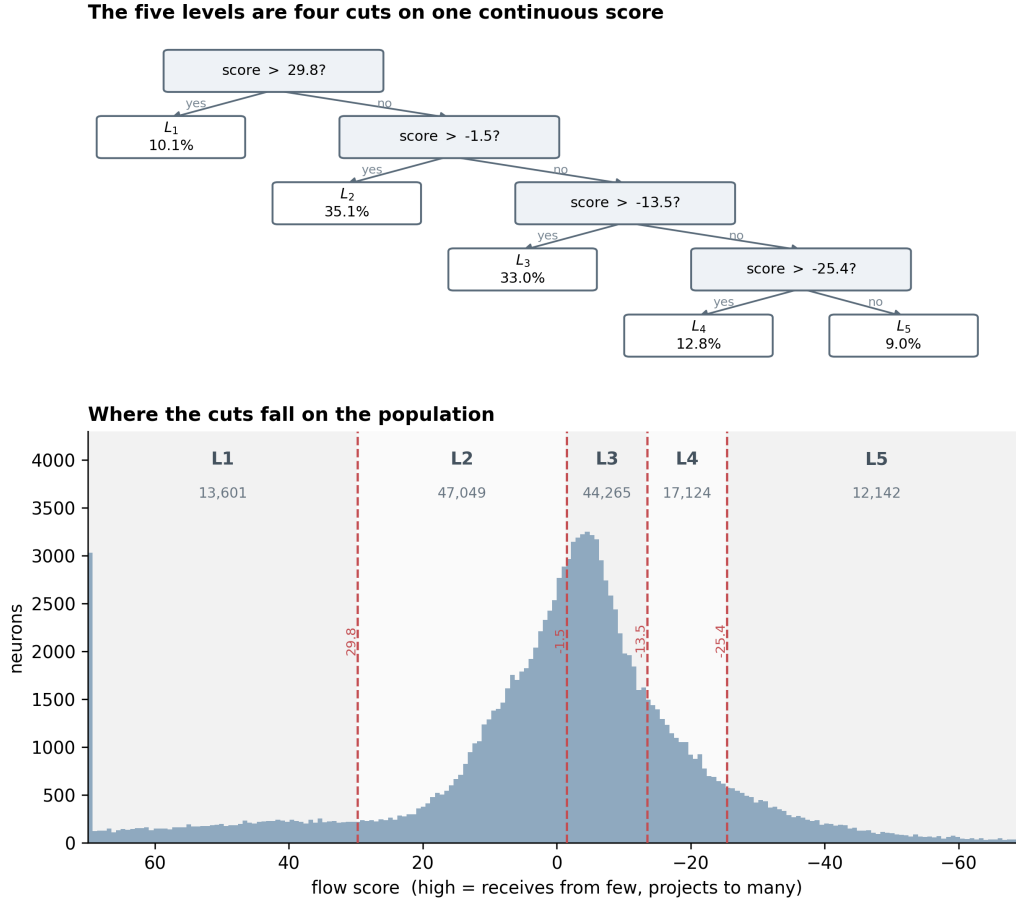


Figure 1: How the five levels are obtained. Above, the decision tree: because all four thresholds apply to the same quantity, the tree is a comb in which each test peels off one level and passes the remainder to the next, and the percentages are the share of the 134,181 neurons falling in each. Below, the same four cuts drawn on the distribution of the flow score itself, with the score increasing to the left so that the sensory end of the axis is on the left and the motor end on the right; the bars at the extremes collect the tails, which run to 906 and -306 . The two central levels are the populous ones, and the cut between them falls almost exactly on the peak of the distribution.

Two things are worth noticing in that figure, because they explain results that appear later. The distribution is heavily concentrated near the middle of the axis, so levels 2 and 3 together hold 68% of the neurons, and the boundary between them cuts through the densest part of the population rather than through a gap. Neither fact is a defect of the construction: a continuous axis has no gaps to cut at, which is precisely why a criterion external to the geometry, namely the super-class, is needed to place the boundaries at all.

This has a consequence that matters for the next section. Since the cut points

were chosen to separate super-classes, levels and super-classes are not independent, and pairing them is in one sense the natural thing to do. This thesis nevertheless groups neurons more finely, for reasons set out in Section 2.2.5, and the cost of that choice is measured rather than assumed.

Once levels are available, every edge acquires a direction relative to the hierarchy. For an edge running from a neuron at level y_u to a neuron at level y_v we write

$$\text{dir}(u, v) = \begin{cases} \text{FWD} & \text{if } y_u < y_v \quad (\text{forward}), \\ \text{BWD} & \text{if } y_u > y_v \quad (\text{backward}), \\ \text{LAT} & \text{if } y_u = y_v \quad (\text{lateral}). \end{cases} \quad (2)$$

A forward edge carries information deeper into the hierarchy, a backward edge returns it towards the periphery, and a lateral edge moves it sideways within a single processing stage. This three-way classification is used throughout the thesis and underlies the finer signature scheme introduced in Section 2.3.

A remark on notation is needed here, because the natural abbreviations are not available. The two directions would ordinarily be written FF for feed-forward and FB for feedback, and this is how they appear in much of the network-motif literature. In this connectome the second of those collides with an anatomical name: FB is the standard abbreviation of the fan-shaped body, a neuropil of the central complex, and it occurs in nineteen of the anatomical groups set out in Section 2.2.4; it also appears in this thesis, in Table 14, as an area rather than as a direction. A reader who meets BWD→FWD without the glossary at hand has no way of telling which of the two is meant. The abbreviations used for directions are therefore FWD, BWD and LAT, which collide with nothing, and the anatomical codes keep their usual meaning. In running text the three are called forward, backward and lateral; the terms *feed-forward* and *feedback* are kept for the general notions as they appear in the literature, and are not used as names for the classes measured here.

2.2.4 The Anatomical Nomenclature

Every result in this thesis names anatomical structures by abbreviation, and those abbreviations are not arbitrary labels: they follow a system that is worth setting out before any of them appears in a figure, since a reader who does not know it cannot interpret a single pattern.

The insect brain is divided into *neuropils*, dense regions of synaptic connections separated by tracts of fibres, and the standard nomenclature of Ito et al. [7] assigns each one a short uppercase code. FlyWire annotates every neuron with the set

of neuropils in which it *arborises*, that is, in which its branches spread out and make synapses, and this annotation is the **group** field of the dataset. The field has 628 distinct values, which sounds unmanageable until one sees how they are built. There are only 44 distinct neuropils in the annotation, and a group is either a single neuropil, of which there are 44, or a pair of neuropils written with a dot between them, of which there are 584. No group names more than two. So **ME** is a neuron confined to the medulla, **LO** one confined to the lobula, and **ME.LO** a neuron that arborises in both and therefore connects them; **AL.MB_CA** spans the antennal lobe and the mushroom body calyx, which is the anatomical signature of an olfactory *projection neuron*, a neuron whose branches span two regions and which therefore carries signals from one to the other. Reading a group name is thus a matter of reading at most two codes, and the dot should be understood as “and”, not as a hierarchy.

Three of these neuropils deserve a word on their own, because they carry most of the results reported later. The *medulla*, the *lobula* and the *lobula plate* are three successive stages of the optic lobe: light is transduced in the retina, passes to the lamina and then to the medulla, which is the largest neuropil in the brain and performs the first substantial visual computation; from there information splits into two parallel streams, one reaching the lobula, associated with the analysis of form and features, and one reaching the lobula plate, which is the stage specialised for detecting motion. A pattern such as **ME.LO** therefore names a neuron bridging two consecutive stages of this pathway, and a circuit running from medulla groups through the lobula to the lobula plate is a circuit moving from early visual processing into motion analysis. The olfactory counterpart is the chain from the *antennal lobe*, which receives input from the smell receptors, to the *mushroom body*, the centre for olfactory learning, and the *lateral horn*, which mediates innate rather than learned responses to odours.

The 44 neuropils are themselves organised into thirteen *super-regions* in the same nomenclature, and this grouping is the macro-level of the anatomy. The medulla, lobula, lobula plate, lamina and accessory medulla make up the optic lobe; the fan-shaped body, ellipsoid body, protocerebral bridge and noduli make up the central complex; the calyx, pedunculus and the two lobes make up the mushroom body; and so on through the superior, inferior, ventrolateral, ventromedial and periesophageal neuropils. Table 1 gives the complete list with the population of each, grouped by super-region. The super-regions matter here for two reasons: they let a reader place an unfamiliar code without consulting an atlas, and they define a third granularity, intermediate between the nine functional super-classes and the 628 groups, which

Table 1: The 44 neuropils that compose the **group** annotation, grouped by super-region and ordered by population within each. The counts divide a neuron between the two neuropils of a composite group, so they sum to the 134,181 neurons that carry both a group and a level. Any group name in this thesis is one of these codes or two of them joined by a dot.

Code	Name	Neurons	Code	Name	Neurons
<i>Optic Lobe</i>			<i>Superior Neuropils</i>		
ME	Medulla	55,811	SMP	Superior Medial Protocerebrum	3,458
LO	Lobula	18,231	SLP	Superior Lateral Protocerebrum	3,150
LOP	Lobula Plate	8,669	SIP	Superior Intermediate Protocerebrum	443
LA	Lamina	7,157	<i>Inferior Neuropils</i>		
AME	Accessory Medulla	36	ICL	Inferior Clamp	980
<i>Ocellar Ganglion</i>			CRE	Crepine	920
OCG	Ocellar Ganglion	274	SCL	Superior Clamp	408
<i>Antennal Lobe</i>			IB	Inferior Bridge	268
AL	Antennal Lobe	3,048	ATL	Antler	65
<i>Lateral Horn</i>			<i>Ventrolateral Neuropils</i>		
LH	Lateral Horn	2,121	AVLP	Anterior Ventrolateral Protocerebrum	3,682
<i>Mushroom Body</i>			PLP	Posterior Lateral Protocerebrum	2,274
MB_CA	Mushroom Body Calyx	2,440	PVLP	Posterior Ventrolateral Protocerebrum	1,607
MB_ML	Mushroom Body Medial Lobe	2,002	AOTU	Anterior Optic Tubercle	926
MB_VL	Mushroom Body Vertical Lobe	484	WED	Wedge	408
MB_PED	Mushroom Body Pedunculus	246	<i>Ventromedial Neuropils</i>		
<i>Central Complex</i>			IPS	Inferior Posterior Slope	1,604
FB	Fan-shaped Body	1,728	SPS	Superior Posterior Slope	1,534
EB	Ellipsoid Body	382	VES	Vest	432
NO	Noduli	376	GOR	Gorget	70
PB	Protocerebral Bridge	108	EPA	Epaulette	10
<i>Lateral Complex</i>			<i>Periesophageal Neuropils</i>		
LAL	Lateral Accessory Lobe	981	SAD	Saddle	968
BU	Bulb	31	PRW	Prow	593
GA	Gall	12	AMMC	Antennal Mechanosensory and Motor Centre	513
			FLA	Flange	334
			CAN	Cantle	8
			<i>Gnathal Ganglia</i>		
			GNG	Gnathal Ganglia	5,379
			<i>Unassigned</i>		
			UNASGD	Unassigned	10

Section 4.5 uses as a control.

The two halves of this description have different standing, and it is worth separating them. The neuropil codes themselves are carried by the data: the per-synapse annotation of the connection table names the same forty-four regions, in hemisphere-resolved form, and the tokens appearing in the **group** field are exactly those. The assignment of neuropils to super-regions, by contrast, is not a field of the dataset at all; it is the hierarchy of the published nomenclature, applied here. Nothing in the data would reveal an error in it, so it is stated as a convention taken from [7] rather than as a measurement.

2.2.5 Group-Level Abstraction

Counting motifs over individual neurons is both computationally hopeless and biologically uninformative. It is hopeless because a five-node motif over 134,000 nodes has astronomically many candidate instantiations, and uninformative because individual neurons are not the units at which circuit function is usually described: what one wants to know is that the antennal lobe projects to the mushroom body calyx, not that one particular neuron does.

We therefore abstract from neurons to *metanodes*. A metanode is a pair consisting of an anatomical category and a hierarchical level, so that the neurons of one structure sitting at level 2 form one metanode and those of the same structure at level 3 form another. A motif is then a pattern over metanodes, and its count is the number of distinct neuron-level instantiations of that pattern. This preserves the quantity that actually matters, namely how many real circuits realise the pattern, while reducing the search space by several orders of magnitude.

Two quantities derived from these counts are used throughout the results and are worth naming now, because they answer different questions and frequently disagree. The *number of patterns* in a set is how many distinct metanode configurations it contains, each counted once however often it occurs. The *mass* of that set is the sum of the counts of its patterns, that is, the total number of neuron-level circuits realising any of them. A class of motifs may contain very many distinct patterns that are each rare, giving it a large number of patterns and a small mass, or a handful of patterns that are each realised enormously often, giving the reverse. Both are reported wherever the two diverge, and percentages are stated as shares of one or the other, never of an unspecified total.

Which categorisation to use is a real choice, and three are available.

The coarsest is the *super-class*, a functional label carried by the FlyWire classification and taking nine values. Since these names recur throughout the results, and since three of them concern the visual system specifically and are easy to confuse, they are set out here. *Sensory* neurons carry signals into the brain from the sense organs. *Optic* neurons lie within the optic lobe, the large visual region that occupies most of each side of the head, and process visual input locally. *Visual projection* neurons carry visual information *out* of the optic lobe towards the central brain, and *visual centrifugal* neurons run the other way, carrying signals from the central brain back *into* the optic lobe. *Central* neurons lie in the central brain and are neither sensory nor motor: this is where association, learning and decision-making happen. *Descending* neurons carry commands out of the brain towards the nerve cord that drives the body, *ascending* neurons carry signals up from the nerve cord

into the brain, *motor* neurons drive muscles directly, and *endocrine* neurons release hormones rather than acting on a specific target. The finest is the *anatomical group* of Section 2.2.4, which takes 628 values. Between them sits the *super-region*, obtained by replacing each neuropil of a group with the super-region that contains it, so that ME.LO becomes OL, both of its neuropils lying in the optic lobe, while LO.PLP becomes OL.VLNP, its two neuropils lying in different super-regions; this yields 79 values.

Paired with the five levels, the three give metanode sets that differ by almost two orders of magnitude. On the shallow window they give 24, 188 and 879 metanodes respectively, with median populations of 511, 11 and 3 neurons and metagraphs of 293, 6,295 and 24,914 edges. Figure 2 shows the coarsest graph in full and the distribution of metanode sizes.

This thesis uses the *anatomical group*. The reason is that the question being asked is which *structure* occupies the waist, and the super-class label cannot express an answer to it: the category *central* alone collapses 521 distinct anatomical groups into one node. The antennal lobe projection neurons, which carry odour information from the antennal lobe to the mushroom body; the Kenyon cells, which are the numerous small neurons forming the body of the mushroom body itself and on which odour memories are stored; and the mushroom body output neurons, which read that structure out towards behaviour, are all, without exception, *central*; so is the lateral horn. Any statement about how information is routed within the central brain is therefore unavailable at that granularity, by construction rather than by accident.

The cost of the finer choice is a long tail of very small categories. The median group holds eight neurons across the whole connectome and the median metanode three on the shallow window, and 344 of the 628 groups hold fewer than ten neurons each. Stated that way the abstraction sounds far too specific to support any general claim, which is why it is worth stating the other half of the same distribution: those 344 small groups together account for 0.8% of the neurons, and 99.2% of the population lives in groups of ten neurons or more. The median is low because the tail is long, not because the analysis rests on tiny categories. The *min_count* threshold of Section 3.3 then removes any pattern supported by fewer than ten instances, so the small categories that do exist are also the ones least likely to survive into a reported result. What remains a genuine limitation is that no threshold can make a category informative if it was small to begin with, and Section 4.5 repeats the entire analysis at both of the coarser granularities in order to measure what that costs.

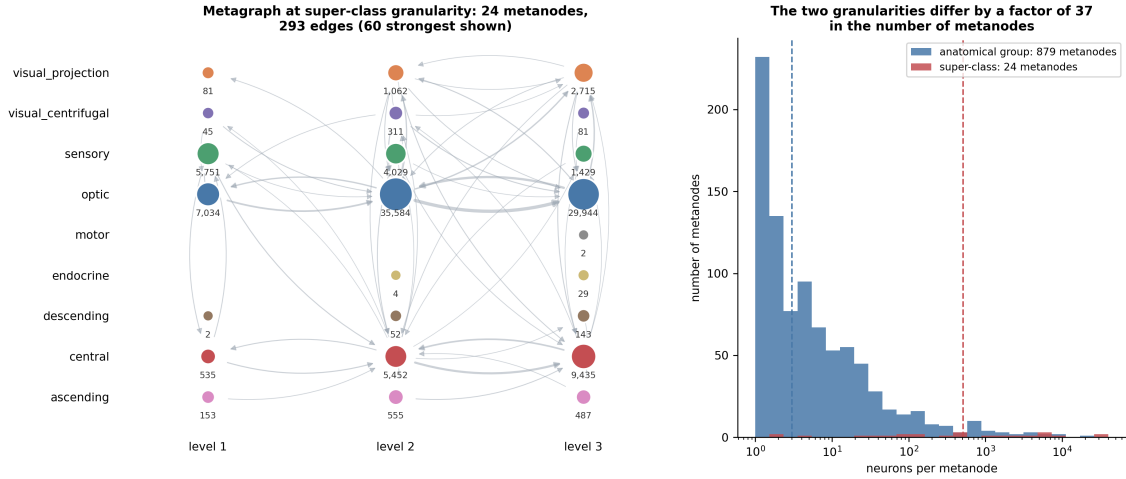


Figure 2: The metagraph on which motifs are counted. On the left, the graph at super-class granularity for window $\{1, 2, 3\}$: each node is a metanode, that is a (super-class, level) pair, placed by level on the horizontal axis and sized by the number of neurons it contains; the sixty strongest of its 293 edges are drawn. On the right, the distribution of metanode populations under the two categorisations, with medians marked. The group-level metagraph has 879 nodes and 24,914 edges and cannot be drawn legibly, which is itself an indication of the difference in resolution between the two.

2.3 Network Motifs

A motif is a small connectivity pattern that occurs more often than chance would predict. This section defines the idea, explains why the comparison against chance is part of the definition rather than an afterthought, and introduces the scheme by which the patterns of this thesis are classified according to the direction their edges run in the hierarchy. That classification is used continuously from here on, so it is developed in full here rather than where the results first need it.

2.3.1 Motifs as Building Blocks

Milo et al. [10] introduced network motifs as patterns of interconnection that recur in a network significantly more often than they do in randomised networks preserving specified properties. The significance requirement is what distinguishes a motif from a merely frequent subgraph, and it is essential: a pattern may be common simply because the network is dense, or because certain nodes happen to have very high degree, without that frequency reflecting anything about the organisation of the system. The comparison against a randomised ensemble is therefore not a statistical formality added at the end but part of the definition, and Section 3.4 describes the three ensembles used here.

2.3.2 Directional Classification

Given the level assignment, every edge of a motif carries one of the three directions of Equation (2). A bow-tie motif with two input groups and two output groups has four edges, two entering the waist and two leaving it. Rather than labelling the motif by all four edges individually, which would produce an unwieldy number of categories, we classify each *branch* by whether its edges agree with one another. If both edges entering the waist are forward, the input branch is labelled **FWD**; if they disagree, it is labelled *mix*. The same applies to the output branch. The resulting label is written

$$\text{signature} = \text{sig}_{\text{in}} \rightarrow \text{sig}_{\text{out}}, \quad \text{sig} \in \{\text{FWD}, \text{BWD}, \text{LAT}, \text{mix}\}, \quad (3)$$

which yields sixteen possible signatures. A motif labelled **FWD**→**LAT**, for instance, is one in which two groups feed forward onto a waist that then distributes laterally within its own level. Figure 3 draws all sixteen.

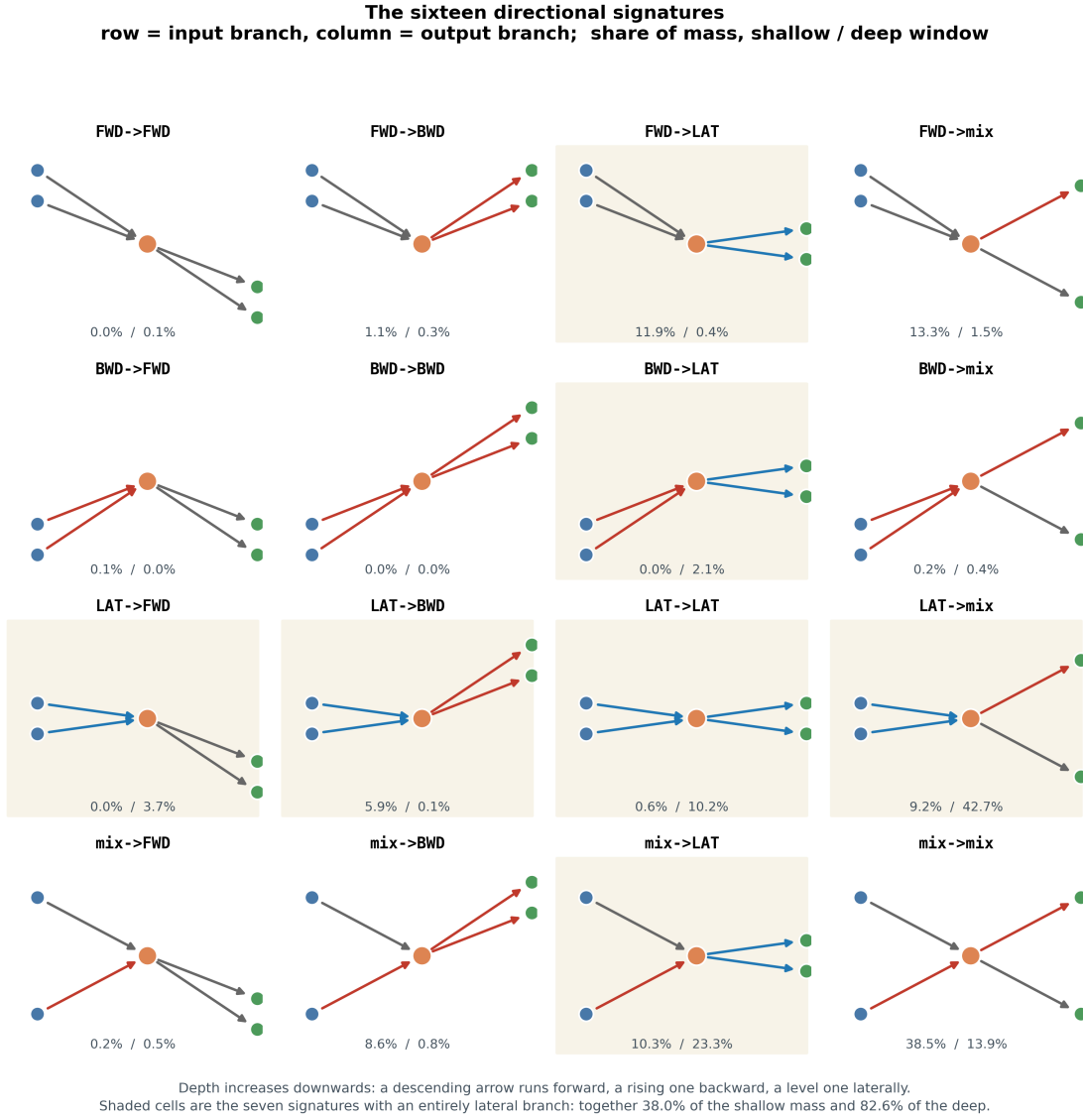


Figure 3: The sixteen signatures, laid out as a matrix: the row gives the direction of the input branch and the column that of the output branch. Depth increases downwards, so the direction of a branch can be read from the slope of its arrows, and a *mix* branch is drawn with one arrow of each kind. Beneath each cell are the shares of motif mass in the shallow and deep windows, from Table 7. The shaded cells are the seven signatures in which at least one branch is entirely lateral; the contrast between their two numbers is the shift reported in Section 4.4.2.

It is worth being explicit about why the signature is defined at this level of detail rather than more simply, because the alternative was tried first and does not work.

The obvious classification, and the one used in earlier work on this dataset, labels a motif by the agreement of all four of its edges at once: a motif is *pure_FWD*, *pure_BWD* or *pure_LAT* when every edge runs the same way, and *mixed* otherwise. The difficulty is arithmetic. Four edges each running in one of three directions give

81 possible combinations, of which only three have all four in agreement, so complete agreement is improbable before one looks at any data at all. On the shallow window the outcome is what that arithmetic predicts: 29,453,022 patterns, 94.4% of the total, fall into *mixed*, while *pure_LAT* takes 1,651,892 (5.3%), *pure_FWD* 57,821 (0.19%) and *pure_BWD* only 22,065 (0.07%). A label that assigns nineteen patterns in twenty to a single class is not telling us much about the connectome; it is telling us that four independent things rarely agree.

The signature of Equation (3) avoids this by asking for agreement within each branch separately rather than across the motif as a whole. Two edges agree far more often than four, so the classes it produces are populated rather than empty, and the thirteen classes hiding inside *mixed* turn out to behave very differently from one another. This is why the finer scheme is used throughout, and why the coarse label appears in this thesis only where a comparison with earlier work requires it.

2.4 Bow-Tie and Hourglass Architecture

This section defines the two structures the thesis is about and separates them, because the literature often does not. The bow-tie is a local pattern and is defined precisely enough to be counted; the hourglass is a global property of a network and is defined precisely enough to be measured. The distinction is what makes it meaningful to ask, later, whether the two agree.

2.4.1 The Bow-Tie Motif

Definition 2.1 (Bow-tie motif). Given metanodes $A_1, \dots, A_{k_{\text{in}}}$, a waist metanode B , and metanodes $D_1, \dots, D_{k_{\text{out}}}$, with $\{A_m\} \cap \{D_n\} = \emptyset$ and $B \notin \{A_m\} \cup \{D_n\}$, a bow-tie motif instance is a tuple of distinct neurons $(a_1, \dots, a_{k_{\text{in}}}, b, d_1, \dots, d_{k_{\text{out}}})$ with $a_m \in A_m$, $b \in B$, $d_n \in D_n$, such that $(a_m, b) \in E$ for all m and $(b, d_n) \in E$ for all n .

Figure 4 shows the structure schematically. The requirement that the input and output groups be disjoint deserves particular emphasis, because it is easy to omit and expensive to omit. Without it a metanode may appear simultaneously among the inputs and the outputs of the same motif, so that what is counted is not a bow-tie at all but a configuration in which a group feeds back into itself through the waist. Such configurations are extremely numerous, because the same group reappearing on both sides multiplies the available combinations, and they are also degenerate in the sense that they do not describe convergence followed by divergence. Enforcing the constraint removes 6.60% of the patterns but 57.3% of the total count mass, which

gives a measure of how strongly degenerate configurations dominate when the search is left unconstrained.

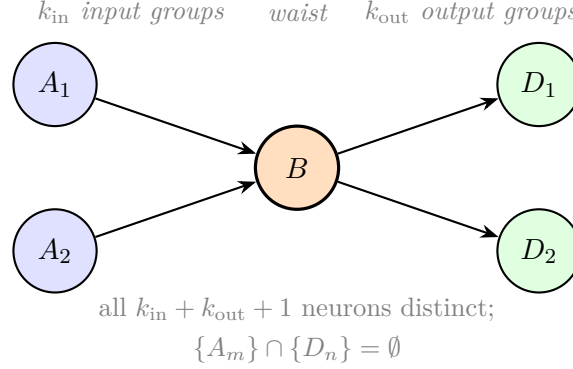


Figure 4: The bow-tie motif of Definition 2.1. Each node is a metanode, that is, a pair consisting of an anatomical group and a hierarchical level; an instance of the motif is a tuple of distinct neurons, one drawn from each metanode, connected as shown. The disjointness requirement between input and output groups is what prevents degenerate configurations in which a group feeds back into itself through the waist.

2.4.2 The Hourglass Effect

Sabrin et al. [11] formalise the hourglass property of a hierarchical network in three steps. One begins by fixing a set S of source nodes and a set T of target nodes; in a connectome these are naturally the sensory and the motor neurons respectively. For each node v of the network one then computes its *path centrality*, defined as the number of source-to-target paths that traverse v . Ordering all nodes by decreasing path centrality and taking the smallest prefix of that ordering which together covers a fraction τ of all source-to-target paths yields the τ -core, written $C(\tau)$. Intuitively, the τ -core is the smallest set of neurons through which most of the traffic must pass.

A small core is not by itself evidence of an hourglass, and this is the point at which the framework becomes subtle. A network might have a small core simply because its source-to-target dependencies are themselves concentrated, quite apart from any hierarchical organisation. The comparison is therefore made against the *flat dependency network*, constructed by connecting every source directly to every target it can reach and discarding all intermediate structure. This flat network has exactly the same dependencies as the original but no hierarchy whatsoever. Writing $C_f(\tau)$ for its core, the hourglass score is

$$H = 1 - \frac{|C(\tau)|}{|C_f(\tau)|}. \quad (4)$$

A value of H close to 1 means that the real network routes its flow through far fewer intermediaries than its dependency structure requires, which is the signature of an hourglass. A value close to 0 means that the apparent concentration is entirely explained by the dependencies themselves and that the hierarchy contributes nothing.

2.5 Previous Work

Two recent theses prepared under the same supervision analyse the same connectome with complementary aims. Tricoli [12] studies neuronal assemblies through motif-based characterisation combined with hierarchical flow ranking, with the goal of identifying groups of neurons that share structural roles and might therefore act as functional units. Curcio [6] introduces a parametric framework for motif discovery conditioned on hierarchical position, and explores the resulting search space along three axes: the vertical depth of the window, the horizontal complexity of the pattern, and the distance spanned by skip connections.

Both of those works enumerate motifs of fixed size and interpret them in terms of super-class composition. The present thesis differs from them in three respects. It targets one specific architectural template, the bow-tie, rather than attempting a full motif census, and this restriction is what makes exact counting through a closed formulation possible where enumeration would not be. It pairs the local analysis with a global hourglass measurement and then quantifies the agreement between the two rather than reporting either in isolation. And it treats statistical validation and the fate of unconfirmed hypotheses as first-class outputs, on the view that an analysis whose controls are not reported cannot be assessed by a reader.

Earlier work on the larval connectome of the same species [5] treated the brain as a flat homogeneous graph and enumerated motifs without conditioning on hierarchical position at all, which produces useful global statistics but collapses together circuits operating at very different processing stages. The present work uses the adult connectome, in which the number of neurons is larger by an order of magnitude, so the contrast is one of method rather than a direct comparison of results.

3 Methods

The two questions of the previous chapter are answered at two different scales, and this chapter describes the machinery for both, in the order in which the results are later presented: the global question first, then the local one.

The order matters and is worth justifying. Asking which structures form waists only makes sense once one knows whether the network has an hourglass organisation

at all; and a measurement of that global organisation is interpretable on its own, whereas a catalogue of local patterns is not, since any large network contains many patterns of any given shape. The global analysis therefore comes first and sets the terms, the local one then identifies the structures responsible, and Section 4.8 compares what the two say about the same tissue.

Both rest on the same foundation, which is why it is described before either. Section 3.1 develops a way of counting how many times a pattern occurs without ever enumerating its occurrences, by exploiting the fact that a count over a branching pattern factorises into a product over the point where the branches meet. That single idea is applied twice: to the waist of a bow-tie, where it gives the number of local patterns, and to a node of the whole network, where it gives the number of sensory-to-motor paths through it. The chapter then covers the statistical validation that distinguishes a motif from a merely frequent pattern, the engineering that makes an exhaustive count feasible, and the checks that establish that the counts are right.

3.1 Counting Without Enumeration

Everything in this thesis rests on being able to count occurrences of a pattern without listing them, because at connectome scale the occurrences number in the hundreds of billions and listing them is not an option. This section develops that capability in four steps: it recasts pattern counting as a linear-algebraic operation, extends it from chains to branching patterns, establishes exactly which patterns the extension covers, and verifies the resulting formulae against brute-force enumeration. The fourth step also fixes the limits of the method, which are as important to state as its reach.

3.1.1 Homomorphism Counting and Flow Vectors

Counting the occurrences of a pattern P in a graph G amounts to counting the homomorphisms from P into G that respect the prescribed group memberships. A homomorphism here is simply an assignment of a neuron to each node of the pattern such that every edge of the pattern is realised by an edge of the graph. For a pattern that is a *path* this reduces to Equation (1): one propagates a vector along the pattern through a sequence of sparse matrix–vector products, and the total count falls out at the end. The vector being propagated is called the *flow vector*, and its i -th entry has a direct interpretation as the number of partial matches of the pattern that terminate at neuron i .

The propagation of the flow vector along a chain of groups follows the implementation of Curcio [6], from which this part of the code derives; the extension to

branching patterns developed in Section 3.1.2 is specific to the present work.

The flow vector has a property that turns out to be decisive. Because the operation is linear, the count computed over a union of groups equals the sum of the counts computed over the individual groups. Counting the pattern on $A_1 \cup A_2$ therefore requires no more work than counting it on A_1 and on A_2 separately and adding the results. This is what allows a search to consider large collections of groups without the cost growing with the number of combinations of them.

3.1.2 The $\Sigma\Pi$ Formulation

A bow-tie is not a path. It is a tree with a branching point at the waist, and at that branching point the two incoming branches are required to be incident on the *same* neuron. That requirement is what destroys the linearity, because the count of the combined pattern is no longer determined by the counts of its parts: knowing how many neurons of A_1 reach the waist and how many neurons of A_2 reach it does not tell us how many neurons of the waist are reached by both.

The correct formulation must therefore work neuron by neuron across the waist. Let $v_{A_m}[i]$ denote the number of neurons of group A_m that project onto neuron i of the waist, and let $w_{D_n}[i]$ denote the number of neurons of group D_n that receive from that same neuron i . Then the exact count of the bow-tie is

$$c(A_1, \dots, A_{k_{\text{in}}}; B; D_1, \dots, D_{k_{\text{out}}}) = \sum_{i \in B} \left(\prod_{m=1}^{k_{\text{in}}} v_{A_m}[i] \right) \left(\prod_{n=1}^{k_{\text{out}}} w_{D_n}[i] \right). \quad (5)$$

Reading the expression from the inside outwards makes its logic clear. For a fixed waist neuron i , the product over m counts the ways of choosing one input neuron from each input group such that all of them project onto i , and the product over n counts the ways of choosing one output neuron from each output group such that i projects onto all of them. Multiplying the two gives the number of complete bow-ties centred on i . Summing over the waist then aggregates across all possible centres. The products enforce coincidence on a single neuron; the sum aggregates over the waist. It is from this structure that the formulation takes its name, $\Sigma\Pi$, and it is exact: no approximation and no sampling are involved anywhere.

Figure 5 works a small example through in full. The left panel isolates one waist neuron and draws its actual connections, so that the four factors can be read off by counting arrows; the right panel repeats those four numbers for every waist neuron and sums the products. Two things are worth noticing. The arithmetic never constructs a bow-tie: it multiplies four counts per waist neuron, of which there are at most a few thousand, where the bow-ties themselves number in the billions. And

a single zero anywhere in a column removes that neuron from the total, which is the observation the pruning of Section 3.3.2 exploits.

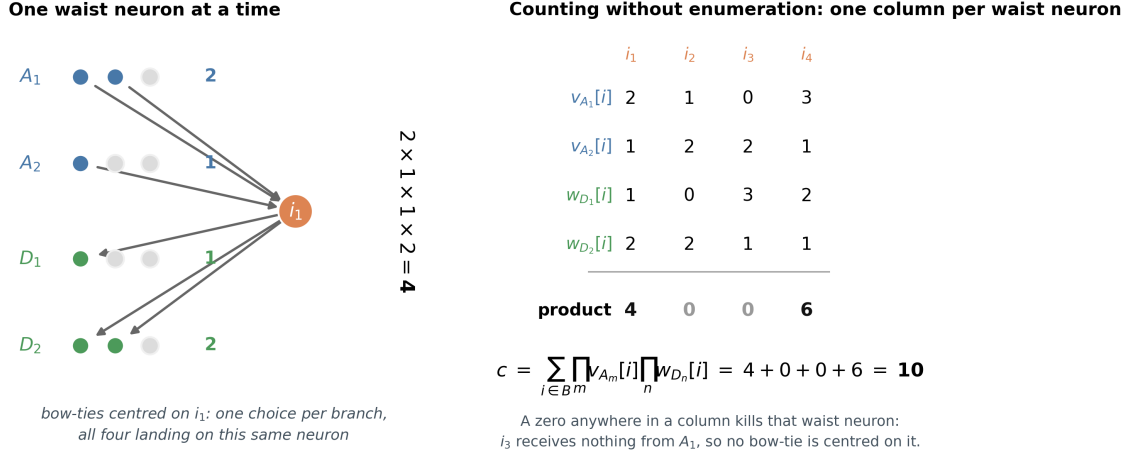


Figure 5: How the count of a bow-tie factorises over the waist. On the left, a single waist neuron i_1 , with the neurons of each group that actually connect to it drawn in colour and the rest greyed out: two neurons of A_1 project onto it, one of A_2 , and it projects onto one neuron of D_1 and two of D_2 , so $2 \times 1 \times 1 \times 2 = 4$ distinct bow-ties are centred on this neuron. On the right, the same four counts for every neuron of the waist, their products, and the sum that gives the count of the pattern. Nothing in the procedure builds an individual bow-tie.

3.1.3 The Algebra of Patterns

The contrast between Equations (1) and (5) is not an accident of the bow-tie. It generalises to a statement about pattern shapes.

Proposition 3.1 (Counting cost by pattern shape). *Let P be a connectivity pattern to be counted in a graph on n nodes. If P is a path, its count is a pure matrix product, computable in $O(n)$ memory, and the count is additive over unions of node groups. If P is a tree, its count requires matrix–vector products along the edges together with an element-wise product at each branching node; memory remains $O(n)$, but additivity fails at the branching nodes. If P is cyclic, its count is not factorisable over vectors at all, and a matrix must be maintained, requiring $O(n^2)$ memory.*

Figure 6 puts the three cases side by side.

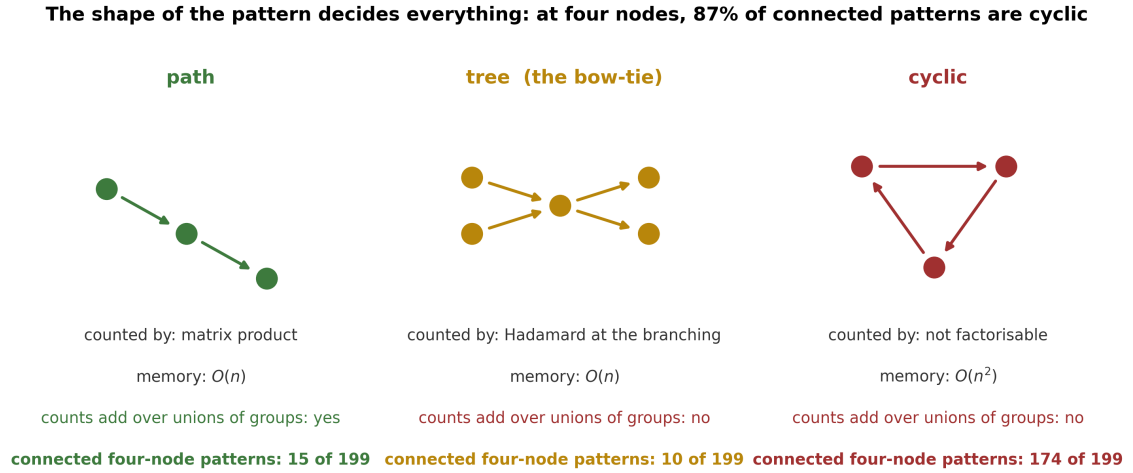


Figure 6: The three classes of Proposition 3.1, with what each costs to count. A path is a chain with no branching: its count is a matrix product and, being linear, it adds across unions of groups. A tree branches but contains no cycle: the branching forces an element-wise product, which keeps the memory linear but destroys additivity, and the bow-tie of this thesis is exactly such a case. A cyclic pattern admits no factorisation over vectors at all. The counts beneath each panel are the connected four-node patterns of each kind, from Table 2.

The practical consequence is that the boundary between the cheap and the expensive is exactly the presence of branching, and that a second boundary, between the feasible and the infeasible at connectome scale, is the presence of a cycle. This is not an empirical observation but a consequence of the structure of the pattern, and it was verified rather than assumed: for all connected patterns up to four nodes, the closed-form counts derived automatically from the pattern were compared against explicit enumeration on random graphs and agreed exactly in every case.

Proposition 3.1 also answers a question that arises naturally when one tries to combine circuits. Given the counts of two linear chains that share a middle node, when is the count of their combination obtained by composing them? The answer is that composition works along the chain and fails at the junction, and the junction is precisely where $\Sigma\Pi$ becomes necessary.

3.1.4 Pragmatic Proof

Proposition 3.1 classifies patterns, but it does not by itself say how much of the space of patterns each class occupies, and without that the classification could be a distinction without practical consequence. To settle the question we enumerated all connected non-isomorphic patterns up to four nodes, derived the factorised counting formula automatically for each of them, and verified every formula against exhaustive

homomorphism enumeration on random graphs.

Table 2: Exhaustive catalogue of connected patterns by shape, with formulae verified against explicit enumeration.

Nodes	Patterns	Paths	Trees	Cyclic	Formulae verified
3	13	6	0	7 (54%)	6/6
4	199	15	10	174 (87%)	25/25

All thirty-one derived formulae reproduced the enumerated counts exactly, which establishes that the automatic derivation is correct on every case it was tested against. The figure worth carrying forward, however, is in the penultimate column of the second row: already at four nodes, 87% of connected patterns are cyclic and therefore not factorisable over vectors. The flow-vector and $\Sigma\Pi$ machinery is thus powerful on the patterns it covers and silent on the large majority of the space. That is exactly why a bow-tie, which is a tree, is a tractable target at connectome scale while a general motif census is not, and it places the present method in relation to enumeration-based approaches rather than in competition with them.

Two boundaries of this result should be stated plainly. The catalogue stops at four nodes because exhaustive enumeration on five would require 2^{20} edge masks for 120 permutations, which this approach cannot reach; the classification itself does not depend on pattern size, so the qualitative conclusion carries, but the percentages beyond four nodes are not measured here. Furthermore, the classification concerns homomorphisms, whereas the requirement that the participating neurons all be distinct is a separate layer sitting on top of the count, handled by *inclusion–exclusion*: one counts without the restriction, then subtracts the cases in which two nodes coincide, then adds back those subtracted twice, and so on. The two should not be conflated: factorisability is a property of the shape of the pattern, whereas node distinctness is a property of the kind of count one wishes to perform.

3.2 Macro-Scale Analysis: the τ -Core

The bow-tie search of the previous section asks a local question: which groups of neurons form convergent–divergent patterns, and how often. It cannot, by construction, answer the global one. A network can be full of local bow-ties without being an hourglass overall, if each of those bow-ties routes a different part of the traffic and none of them is on the path of everything else; and conversely a network could funnel all of its traffic through a handful of neurons that never appear in any frequent

local pattern. The two questions are independent, and the second is the one the architecture literature actually asks.

This section describes how it is answered. The idea, taken from Sabrin and Dovrolis [11], is to stop looking at patterns and start looking at *paths*. If the network really is an hourglass, then a small set of nodes must lie on almost all the routes from the sensory periphery to the motor output, and removing that set must disconnect almost everything. Measuring how small that set is, and comparing it against how small it would have to be in a network with the same inputs and outputs but no intermediate structure at all, gives a single number for how hourglass-like the architecture is. Three definitions are needed to make this precise: the centrality of a node with respect to paths, the minimal set that covers a given fraction of them, and the reference against which its size is judged.

3.2.1 Path Centrality and the τ -Core

Fix a set of sources S and a set of targets T , and let Π be the set of paths from some source to some target. The *path centrality* of a node v is the number of those paths that pass through it,

$$P(v) = \left| \{ \pi \in \Pi : v \in \pi \} \right|. \quad (6)$$

This is computed without enumerating any path. Writing $a_h(v)$ for the number of paths of exactly h hops from a source to v , and $b_h(v)$ for the number of h -hop paths from v to a target, both obtained by propagating along the layers of the routing, the centrality of v under a bound H on path length is

$$P(v) = \sum_{h_1+h_2 \leq H} a_{h_1}(v) b_{h_2}(v), \quad (7)$$

which is the same factorisation over a meeting point that Equation (5) uses for the waist of a bow-tie. The macro and micro analyses therefore rest on the same algebra, applied once to a pattern and once to a whole network.

For a set of nodes X , let $\pi(X)$ denote the fraction of paths in Π that pass through at least one member of X . The τ -core is the smallest set covering a fraction τ of all paths,

$$C(\tau) = \arg \min_{X \subseteq V} |X| \quad \text{subject to} \quad \pi(X) \geq \tau. \quad (8)$$

Finding the smallest such set exactly is an instance of *set cover*, the problem of choosing the fewest sets that between them cover everything, which is NP-hard: no algorithm is known that solves it exactly in time polynomial in the size of the input, and at this scale an exact search is out of the question. The core is therefore built

greedily, as in the original work: at each step the node of highest path centrality is added to the core and its paths are removed, and the centralities are recomputed on what remains. Throughout this thesis $\tau = 0.9$, so the core is the smallest set of neurons that carries nine tenths of the sensory-to-motor traffic.

3.2.2 The Flat Reference and the Hourglass Score

The size of $C(\tau)$ is not interpretable on its own, because it depends on how many sources and targets the network has. A network with ten thousand sensory neurons will need a large core simply to touch all of them, hourglass or not. What is needed is a reference: how large would the core have to be in a network that has the same sources, the same targets and the same source-to-target dependencies, but no intermediate structure through which to share them?

That reference is the *flat dependency network* G_f , which retains only the sources and the targets and joins each pair (s, t) by a single edge whenever a path from s to t exists in the original network. Every path in G_f has length one, so no node can ever lie on the route of another pair, and covering a fraction τ of the dependencies requires naming the sources and targets themselves. Writing $C_f(\tau)$ for the core of the flat network, computed by the same greedy procedure, the *hourglass score* is

$$H = 1 - \frac{|C(\tau)|}{|C_f(\tau)|}, \quad H \in [0, 1). \quad (9)$$

The interpretation follows directly from the definition. If the network routes its traffic through a narrow waist, then $|C(\tau)|$ is far smaller than $|C_f(\tau)|$ and H approaches one. If it has no shared intermediate structure, the two cores are of comparable size and H approaches zero. The score therefore measures not whether a network has a bottleneck in absolute terms, but how much narrower its bottleneck is than the flat network its own inputs and outputs would force. Figure 7 shows the comparison on a sketch small enough to check by eye.

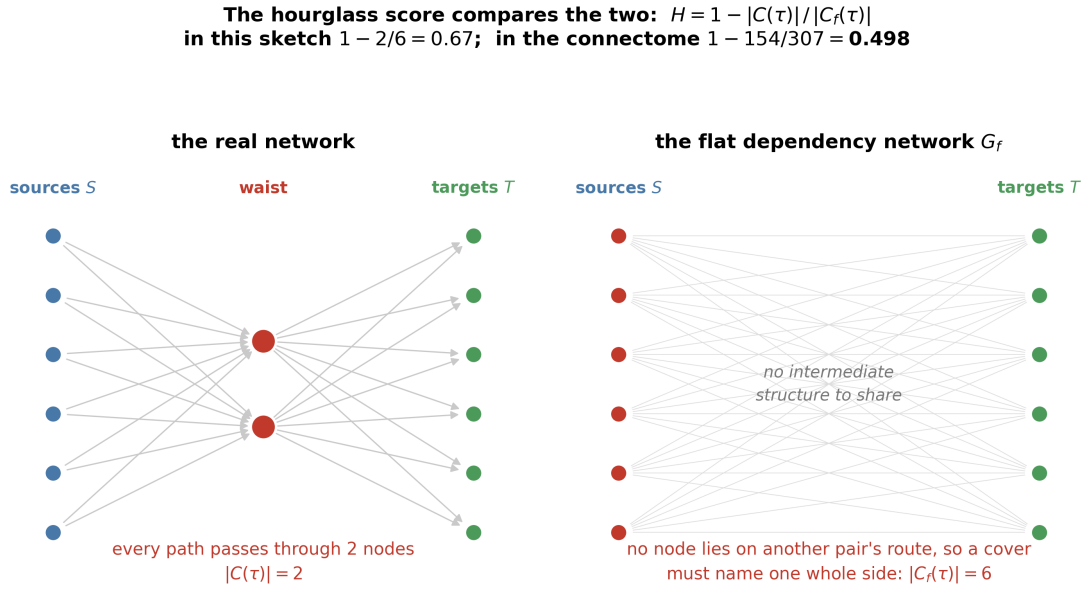


Figure 7: Why the size of the core has to be judged against a reference. On the left, a network in which every route from a source to a target passes through one of two intermediate nodes: naming those two covers everything, so the core has size two. On the right, the flat dependency network built from the same sources and targets, in which each dependent pair is joined directly. No node there lies on any other pair's route, so covering the dependencies requires naming an entire side, six nodes. The hourglass score is one minus the ratio of the two, and it is large exactly when the real network shares intermediate structure that the flat one cannot.

3.2.3 Sources, Targets and Routing

The sources S are taken to be the sensory neurons, of which there are 12,711, and the targets T the motor and descending neurons, of which there are 1,382. Computing path centrality then requires a decision about which paths are to be counted, and this decision matters more than it might appear.

Two routing schemes are used. Under **levels** routing, a path may only traverse edges that do not decrease the hierarchical level. This makes the routing acyclic by construction, so that path counts are finite without any further restriction, at the cost of discarding every lateral edge; because the acyclicity comes from the structure of the hierarchy itself rather than from an imposed bound, this scheme is referred to as *structural routing* wherever the results are discussed. Under **score** routing, paths follow a continuous flow score which recovers part of the lateral connectivity, with an explicit bound imposed on the length of admissible paths. Since almost 39% of all edges in this connectome are lateral, the difference between the two schemes is substantial and both are reported throughout.

3.2.4 Why the Path Length Bound Is Mandatory

The bound H on path length in Equation (7) was introduced without comment, and it deserves one, because it is not a computational convenience but a condition for the centrality to be defined at all.

In a cyclic graph the number of source-to-target paths diverges combinatorially with the permitted path length, since paths may wander through cycles arbitrarily many times. As the count diverges, the ordering of nodes by path centrality degenerates, because a single well-connected neuron appears on an overwhelming proportion of the increasingly baroque paths. Section 4.3 quantifies exactly how bad this becomes. The bound on path length is therefore not a computational convenience that could be relaxed given more resources, but a definitional requirement without which the measure has no content.

3.3 Micro-Scale Analysis: Bow-Tie Motif Search

Where the previous section asked whether the network as a whole funnels its traffic, this one asks which structures do the funnelling. The object counted is the bow-tie of Definition 2.1, over metanodes rather than neurons, and the counting uses the $\Sigma\Pi$ formulation of Section 3.1.2 directly.

Three things have to be settled before a search can be run: which configurations are admissible, how the space of candidates is cut down enough to make an exhaustive pass feasible, and how the resulting waists are compared with one another. This section takes them in that order.

3.3.1 Search Space and Parameters

The search is governed by four parameters. The level window NL fixes the set of hierarchical levels within which motifs are sought, so that $NL = \{1, 2, 3\}$ restricts attention to early sensory processing while $NL = \{3, 4, 5\}$ examines deep integration. The parameters k_{in} and k_{out} set the number of input and output groups of the bow-tie. The parameter *max_jump* bounds the level difference that a single edge of the metagraph may span, and setting it to one imposes a layer-by-layer processing logic in which a connection may move at most one level up or down. Finally *min_count* sets the minimum number of occurrences a pattern must have to be retained, which suppresses the long tail of patterns realised by only a handful of neurons. Unless stated otherwise, all results in this thesis use $k_{\text{in}} = k_{\text{out}} = 2$, *max_jump* = 1 and *min_count* = 10.

The choice $k_{\text{in}} = k_{\text{out}} = 2$ deserves a word, since the framework imposes no such

restriction. It is the smallest configuration that is genuinely a bow-tie: two inputs are the minimum needed for convergence and two outputs the minimum for divergence, whereas $k_{\text{in}} = k_{\text{out}} = 1$ describes a linear chain rather than a waist, and linear chains through this hierarchy have already been characterised in earlier work on the same dataset [6]. Taking the smallest structure that expresses the architecture under study also keeps the count interpretable, since every pattern is then a statement about four edges rather than about an increasingly baroque configuration.

The framework is nonetheless parametric in both values, and the search engine accepts any k_{in} and k_{out} : configurations such as $(3, 3)$ and $(3, 2)$ run without modification. What makes larger values impractical is cost rather than expressiveness. The number of group combinations per waist grows as $\binom{p}{k_{\text{in}}} \binom{q}{k_{\text{out}}}$, so a waist with fifty incoming and fifty outgoing groups goes from 1.5 million combinations at $k_{\text{in}} = k_{\text{out}} = 2$ to 384 million at three, a factor of 256; interpretation degrades in the same direction, since each pattern then constrains six edges rather than four.

A genuine implementation limit applies only to the parameter sweep of Section 4.10. That sweep avoids enumerating combinations by evaluating an *elementary symmetric polynomial*: given a list of numbers, the k -th such polynomial is the sum of all products of k of them taken at a time, which is exactly the quantity a sum over all k -combinations of groups requires, and which can be computed without forming a single combination. That shortcut admits an exact disjointness correction only for the pure forms $(k, 1)$ and $(1, k)$; the general case (k, k) with $k \geq 3$ would require inclusion–exclusion to a depth this implementation does not reach.

3.3.2 Pruning

Three constraints reduce the search space, and it is important that none of them affects exactness: each removes only combinations that provably contribute nothing. The first is the disjointness requirement of Definition 2.1, which excludes combinations in which an input group also appears among the outputs. The second is a restriction to the support of the waist: only those waist neurons that are reachable from at least one input group and that in turn reach at least one output group can possibly contribute to a count, so all others are removed before any product is formed. The third is an early exit on null flow, which discards any combination in which one of the factors of Equation (5) is identically zero, since the product is then zero regardless of the remaining factors. On the shallow window these three together eliminate more than 97% of the nominal search space.

3.3.3 Compression Metrics

A waist is interesting to the extent that it *compresses*, that is, to the extent that many peripheral neurons are routed through few central ones. The obvious way to measure this is to take the ratio of the number of peripheral neurons to the size of the waist group, but that measure fails in practice, and understanding why it fails motivates the correct one.

The difficulty is that a group may be nominally large while only a handful of its neurons actually participate in any given pattern. A waist of twenty neurons of which two carry all the flow behaves, functionally, like a waist of two, and dividing by twenty understates its compression by an order of magnitude; conversely, a group of twenty neurons that all participate equally is a genuinely distributed bottleneck. What is needed is a measure of the *effective* number of participating neurons, one that equals the group size when all neurons contribute equally and falls towards one when a single neuron dominates.

The participation ratio provides exactly that. Writing P_i for the $\Sigma\Pi$ contribution of waist neuron i , that is, the number of bow-ties centred on that neuron, we define

$$n_{\text{waist}}^{\text{eff}} = \frac{(\sum_i P_i)^2}{\sum_i P_i^2}, \quad \text{realised compression} = \frac{\text{peripheral neurons connected to an active waist}}{n_{\text{waist}}^{\text{eff}}}. \quad (10)$$

If k neurons contribute equally and the rest not at all, the numerator is $(kP)^2$ and the denominator kP^2 , so that $n_{\text{waist}}^{\text{eff}} = k$ as required. If one neuron carries everything, both sums are dominated by that neuron and $n_{\text{waist}}^{\text{eff}}$ tends to one. Section 4.6 shows what difference the correction makes to the ranking of waists in practice.

3.4 Statistical Validation

A pattern that occurs frequently is not thereby a motif. Large networks contain enormous numbers of every small pattern simply because they are large and because some neurons have very many connections, and a count on its own cannot distinguish structure from that background. The comparison against randomised networks is therefore not a check applied at the end but part of what the word motif means, as Section 2.3 noted.

The difficulty is that there is no single correct randomisation. Each choice of what to preserve and what to destroy tests a different explanation, and a pattern can be significant against one and unremarkable against another; this thesis reports a case where the ordering of patterns inverts between two of them. Three null models are therefore used rather than one, and this section describes them, the way significance

is assessed given the shape of the resulting distributions, and how the size of an effect is reported once significance is established.

3.4.1 Three Complementary Null Models

A single null model tests a single confound, and can therefore only ever establish that the observed structure is not explained by that one thing. Because several different properties of the connectome could plausibly generate an apparent excess of bow-ties, we use three null models, each of which destroys a different aspect of the observed structure while preserving the others.

The three are the following.

- **Model A, degree-preserving.** The classical null of Milo et al. [10, 9]. The network is rewired through repeated edge swaps that leave the in-degree and out-degree of every neuron unchanged. It preserves the degree sequences exactly and destroys the anatomy completely, so a pattern still over-represented under it cannot be explained by degree alone.
- **Model B, block-density-preserving.** Edges are resampled independently within each pair of anatomical groups, at the density observed for that pair. It preserves how strongly any two regions are connected and destroys individual degrees, so it tests whether a pattern is explained by regional connectivity alone.
- **Model C, degree permutation within blocks.** The strictest of the three. Within each block the assignment of degrees to individual neurons is permuted, preserving both the number of edges and the per-block degree sequences, and randomising only *which* waist neuron receives which degree. It leaves everything intact except the coincidence structure at the waist, which is precisely the property a bow-tie count depends on.

The three are therefore nested in strictness rather than alternative to one another, and a pattern that survives all three has been shown not to follow from degree, from regional density, or from the way degree is distributed within a region. Figure 8 shows what each one does to a small network.

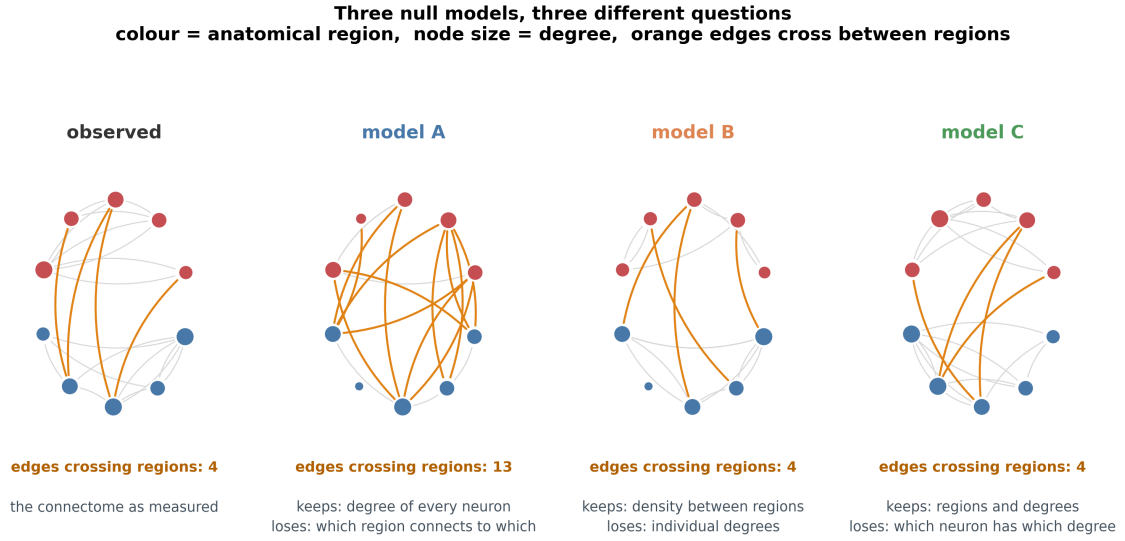


Figure 8: The three null models applied to the same small network, whose ten nodes belong to two anatomical regions shown in two colours. The size of a node is its degree and the edges that run between the two regions are drawn in orange, so the two properties each model is concerned with are both visible. Model A keeps every degree but scatters the edges across regions, and the count of crossing edges rises sharply. Model B keeps the number of crossings and randomises the degrees. Model C keeps both and permutes only which node within a region carries which degree, which is why it is the strictest of the three: what it randomises is precisely the coincidence structure a bow-tie count depends on.

3.4.2 Inference

Significance is assessed with empirical p -values,

$$p = \frac{1 + \#\{c_{\text{rand}} \geq c_{\text{real}}\}}{N + 1}, \quad (11)$$

where c_{real} is the observed count and the c_{rand} are the counts obtained on N randomised networks. We use empirical p -values rather than z -scores because the null distribution of motif counts is strongly right-skewed, so that a z -score does not correspond to a tail probability and would systematically overstate significance.

The set of patterns to be tested is *pre-registered*: it consists of the twenty most frequent patterns of each of the sixteen signatures, three hundred and twenty patterns in all, selected before any randomisation was examined. This matters because selecting patterns after seeing which ones look unusual would guarantee significance by construction. Multiple testing across the three hundred and twenty tests is controlled with the Benjamini–Hochberg false discovery rate procedure [2], which bounds the expected proportion of false positives among the rejections.

Randomisation uses $N = 100$ samples with a fixed seed, and the mixing of the edge-swap chain was verified empirically rather than assumed by checking that the results stabilise well before the number of swaps actually used.

3.4.3 Effect Size

The ratio between the real count and the median randomised count is deliberately *not* reported as an effect size anywhere in this thesis. It spans several orders of magnitude across the three null models and is dominated by how thoroughly each model destroys the structure rather than by any property of the data, which makes it uninterpretable as a measure of biological strength. Section 4.7.1 shows that the ratio does not even preserve the ordering between signatures from one null to another, which settles the matter.

3.5 Computational Realisation

The formulation of Section 3.1 makes an exhaustive count possible in principle. Making it possible in practice, on a network of 134,181 neurons, required three pieces of engineering, and they are described here because the analysis of the deeper windows would not have been feasible without them: a reformulation of the inner loop as a matrix product, a sparse variant for the cases where the dense form does not fit in memory, and a decision about what to keep of an output that would otherwise run to hundreds of gigabytes.

3.5.1 The Matrix Engine

A direct implementation of Equation (5) loops over combinations of groups and performs, for each one, a copy followed by k_{out} multiplications and as many additions over arrays of the length of the waist. On the shallow window this amounts to roughly 1.7 billion iterations, and profiling shows something initially surprising: the cost is dominated not by the arithmetic but by the per-call overhead of the array library, some two microseconds per call. The computation is therefore limited by how often the inner loop crosses into the numerical library, not by how much numerical work it does.

The remedy is to arrange for all combinations of a given waist to be computed at once. Let P_{in} be the matrix whose row r is the element-wise product of the input flow vectors of combination r , and let P_{out} be the analogous matrix for the output

combinations. Then every count belonging to that waist is an entry of

$$T = P_{\text{in}} P_{\text{out}}^{\top}, \quad (12)$$

a single dense matrix product which can be delegated wholesale to the optimised linear algebra library. Billions of individual calls collapse into a few thousand matrix products.

The compression metrics of Section 3.3.3 follow the same pattern and need not be computed separately per combination. Since the per-combination vector is the element-wise product of an input row and an output row, the sum of its squares and the count of its non-zero entries are themselves matrix products,

$$\sum_i p_{v,i}^2 = P_{\text{in}}^2 (P_{\text{out}}^2)^{\top}, \quad n_{\text{active}} = (P_{\text{in}} > 0)(P_{\text{out}} > 0)^{\top}, \quad (13)$$

so that the vector itself is never materialised for any individual combination.

3.5.2 The Sparse Engine

Equation (12) requires memory proportional to the number of combinations multiplied by the size of the waist, and therefore switches itself off precisely on the largest waists. This is unfortunate, because those are the expensive ones: on the shallow window the waists that exceed the memory budget account for the majority of the total runtime.

The way out comes from observing that on a large waist the flow vectors are almost entirely zero. An input group typically reaches only a few hundred of the 24,835 neurons of the largest metanode, so the matrices in Equation (12) are dense representations of almost nothing. Computing the same product with sparse operands produces directly, and only, the pairs whose count is non-zero, and those are exactly the pairs that need to be emitted. Memory then scales with the number of non-zero entries rather than with the product of the dimensions, and the constraint that forced the fallback disappears. The three quantities computed alongside the count share the same sparsity structure, since the values are non-negative counts and a product vanishes if and only if one of its factors does; that this alignment holds is asserted at run time rather than assumed.

3.5.3 Output Distillation

Exhaustive counting produces one row of output per pattern, and for the full window that means 512,969,374 rows. Retained as plain text these rows occupy more than

100 GB across the four windows, which is unwieldy to store and slow to read.

Examining what the downstream analyses actually consume shows that none of them uses the individual rows. Every one of them either aggregates the rows into summary counts or selects a top- N subset by frequency. The row-level output is therefore an intermediate rather than a result: it needs to exist for the duration of a single pass and no longer. The search engine consequently retains only the aggregates, computed by signature, by waist and by level chain, together with top- N selections in which the complete rows are kept. This reduces the four windows from over 100 GB to 68 MB while producing downstream results that are identical.

The selections are deliberately wider than any current analysis requires, and they are stratified by waist and signature jointly as well as by waist alone, so that comparisons made at parity of waist between classes of motif remain possible. Section 4.9 depends on exactly that stratification.

3.6 Correctness Verification

Three independent mechanisms are used to establish correctness, and they are described here in some detail because the credibility of every number in Section 4 rests on them.

The first is a reference implementation. A slow but readable implementation of Equation (5), written directly from the mathematics without any of the optimisations, is retained alongside the fast engine. The two are compared row by row across all 1,043,872 rows and all 26 columns of a complete run, including 24 waists deliberately forced onto the sparse path by lowering the memory budget so that the sparse code is exercised rather than merely present.

The second is exhaustive enumeration. Every closed-form count in this thesis is checked against explicit enumeration on random graphs where enumeration is feasible: the within-area formula of Section 4.11 on three hundred random graphs, the symmetric-polynomial formulation used for the $(k_{\text{in}}, k_{\text{out}})$ sweep, and the pattern algebra of Proposition 3.1. A closed form that agrees with brute force on small cases can still be wrong on large ones, but a closed form that disagrees on small cases is certainly wrong, and this check has caught real errors during development.

The third is internal consistency. Each analysis recomputes the occurrence count of every pattern it handles directly from the graph and compares it against the stored value, so that a corrupted or misaligned intermediate would be detected at the point of use. All such checks reported exact agreement.

4 Results

The results follow the same path as the methods: from the whole network inwards.

They open with the hierarchy on which everything else is conditioned, establishing what the five levels contain and how densely they are wired, and then testing whether that hierarchy remains meaningful when neurons are grouped by anatomy rather than by function. With the axis established, the global question is asked: does this connectome route its traffic through a narrow waist, and how does the answer compare with the one obtained for *C. elegans*. The analysis then descends to the local scale, where the exhaustive motif search across four level windows identifies which structures actually form those waists, how the picture changes with depth, and how much the choice of anatomical granularity determines what can be seen. The two scales are then compared directly against each other.

The remaining sections take up questions that either scale raises on its own: whether the counts survive randomisation, what the neurochemistry of the waists is, how the picture changes when the shape of the pattern is allowed to vary from integrative to broadcast, what happens inside a single anatomical area, and what the whole analysis costs to run.

4.1 Hierarchical Structure of the Connectome

Everything that follows is conditioned on the five hierarchical levels, so the first thing to establish is what they contain and how they are wired to one another. This section reports the size and functional composition of each level, the density of connections between them, and the balance of forward, backward and lateral edges. The last of these turns out to matter for the macro-scale analysis, which discards a substantial fraction of the edges depending on how paths are routed.

4.1.1 Level Sizes and Composition

Since the level assignment is inherited rather than derived here, the first thing to establish is that it means what it is supposed to mean. Table 3 reports how the neurons are distributed across the five levels together with the functional composition of each level, and Figure 9 shows the same two quantities.

Table 3: Neurons per hierarchical level and functional super-class composition (percentages by row).

Level	Neurons	%	optic	sensory	central	descending	vis. proj.
L1	13,601	10.1	51.7	42.3	3.9	0.0	0.6
L2	47,049	35.1	75.6	8.6	11.6	0.1	2.3
L3	44,265	33.0	67.6	3.2	21.3	0.3	6.1
L4	17,124	12.8	26.9	4.9	48.0	1.4	16.3
L5	12,142	9.0	3.2	5.5	71.3	6.9	8.3

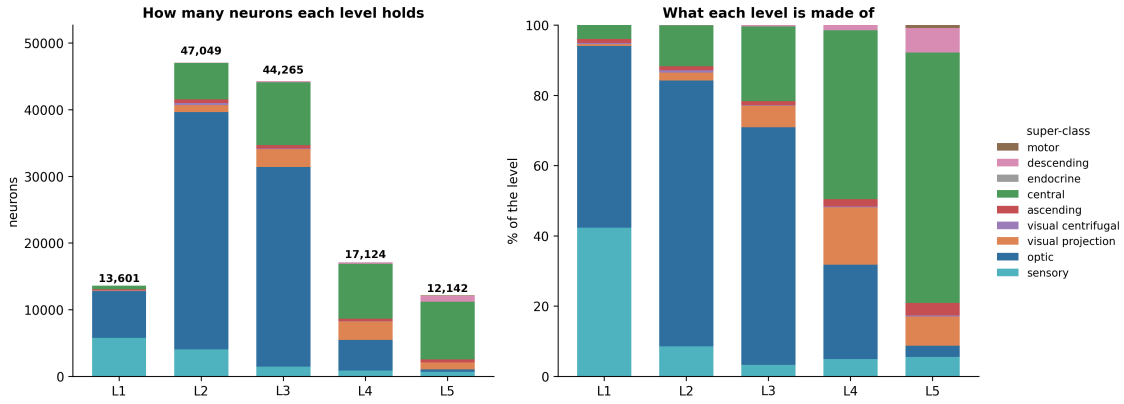


Figure 9: The five levels by size and by content. On the left, how many neurons each level holds; on the right, the same populations normalised, so that the composition can be compared across levels of very different size. The super-classes are stacked in sensory-to-motor order, and the resulting picture is a gradient: sensory and optic neurons fill the shallow levels, central neurons take over from level 4, and descending neurons, which carry signals out of the brain towards the motor system, appear in quantity only at level 5.

The gradient is monotone in the expected direction. Sensory content falls from 42.3% at level 1 to 5.5% at level 5, central content rises from 3.9% to 71.3% over the same range, and descending neurons appear only at the two deepest levels.

How much this confirms requires care, because the level boundaries were themselves chosen to separate super-classes, as Section 2.2.3 described. Reading the gradient as independent confirmation of the levels would therefore be circular, and it is not claimed here. What the gradient does confirm is something else, and it is worth separating the two.

The construction has two stages, and only the second one saw the super-class labels. The spectral embedding that produces the flow score is computed from connectivity alone: it knows which neuron projects to which, and nothing about what

any of them does. The decision tree then chose four numbers on that one-dimensional axis. Four thresholds are very little freedom, and no choice of them could have produced the gradient of Figure 9 if the underlying axis had not already ordered the neurons that way. The alignment is therefore evidence about the *embedding*, not about the discretisation: a quantity derived purely from wiring turns out to sort neurons into the same order that their biological function does. That is a genuine result, and it is the reason the levels can be trusted as an axis of information flow. What it is not is a validation of where exactly the four cuts fall, which remains a modelling choice inherited from earlier work.

4.1.2 Connection Density

Raw counts of edges between levels are dominated by the sizes of the levels involved: level 2 alone contains 47,049 neurons, so any transition touching it appears large simply on that account. The informative quantity is therefore the density, defined as the number of edges observed per ten thousand pairs of neurons that could possibly be connected. Figure 10 shows the full matrix.

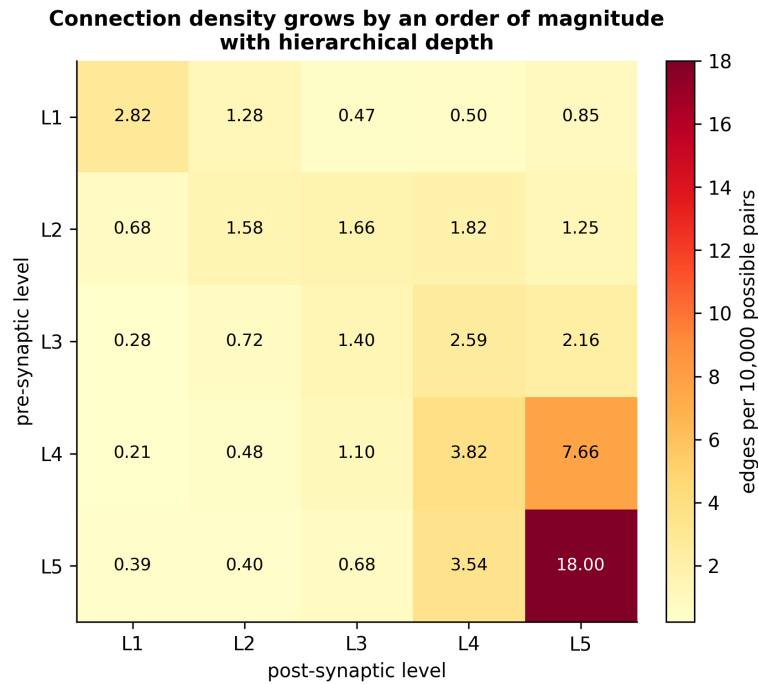


Figure 10: Connection density between hierarchical levels, expressed as edges per ten thousand possible pairs. The normalisation removes the effect of level size, which dominates raw edge counts. Density grows by an order of magnitude with depth, from 1.28 along L1→L2 to 18.00 within L5.

Density grows by an order of magnitude with depth, from 1.28 along the L1-to-L2

transition to 18.00 within level 5 itself. The deep brain is therefore not a smaller copy of the shallow brain but a qualitatively denser structure, in which each neuron participates in far more connections relative to the population available to it. This single observation recurs throughout the results and explains several of them.

4.1.3 Edge Direction

Classifying every edge of the connectome by Equation (2) gives 1,169,787 forward edges, 1,054,005 lateral edges and 476,721 backward edges. Almost 39% of all connections are therefore lateral, running between neurons of the same hierarchical level. This has an immediate methodological consequence: any routing scheme that discards lateral edges is discarding a very substantial fraction of the connectome, which is why Section 4.3 reports both schemes rather than choosing one.

4.2 Do the Levels Survive the Change of Granularity?

The hierarchy of Section 2.2.3 was built in two steps, and the second of them deserves scrutiny before the levels are used any further. A spectral embedding placed every neuron on a continuous axis, and a decision tree then cut that axis into five bins, choosing the cut points so as to separate the functional super-classes as cleanly as possible. The levels are therefore calibrated against the super-class, whereas everything in this thesis pairs them with the anatomical group instead. Nothing guarantees in advance that a construction tuned for one categorisation remains meaningful under another, and the question has to be settled with a measurement rather than an assurance.

The measurement is simple to state. For each category one takes the distribution of its neurons across the five levels and asks how concentrated it is. Three quantities summarise this: the *purity*, which is the share of the category's neurons falling in its most populated level; the *normalised mutual information* between category and level, which is the amount the category tells us about the level divided by the total uncertainty of the level and is therefore comparable across categorisations of different sizes; and the *contiguity*, which asks whether the levels a category actually occupies form a consecutive run rather than a scattering. Each is compared against a reference in which the level labels are shuffled at random between neurons, preserving both the sizes of the categories and the overall distribution of levels and destroying only the association between the two.

The first thing the measurement shows is that the concern is well founded as stated. Anatomical groups are not confined to a single level: their mean purity,

weighted by population, is 0.563, and only 179 of the 628 groups live entirely on one level. The lobula plate is a clear case, with 647 of its neurons at level 2, 1,077 at level 3 and 157 at level 4; the lobula is more dispersed still, with a modal share of only 0.452. Whatever else is true, it is not true that a group determines a level.

Table 4: Coherence between anatomical category and hierarchical level, at the three granularities. Purity is the share of a category’s neurons in its modal level, averaged with weights proportional to category size; NMI is $I(\text{category}; \text{level})/H(\text{level})$. The random column shuffles the level labels between neurons, preserving category sizes and the level distribution.

Granularity	Categories	Purity	Random	NMI	Random
super-class	9	0.413	0.351	0.144	0.000
super-region	79	0.461	0.352	0.190	0.001
anatomical group	628	0.563	0.356	0.301	0.006

The second thing it shows is that the implication does not follow, and it does not follow for three separate reasons.

The first is that a metanode is a *pair*, not a group. The unit over which motifs are counted is (group, level), so LOP_L2, LOP_L3 and LOP_L4 are three different nodes of the metagraph and are never merged. Dispersion of a group across levels is therefore not an error the abstraction has to absorb: it is the reason the abstraction carries the level at all. Were every group confined to one level, the level would be a deterministic function of the group and would add nothing whatever; the pairing is informative precisely to the extent that groups are spread. The measured dispersion is what makes the second coordinate worth having.

The second is that the dispersion is structured rather than diffuse. Of the 628 groups, 570 occupy a consecutive run of levels once levels holding less than five per cent of the group are discarded, and those 570 groups contain 99.4% of all neurons. A group does not scatter over the hierarchy; it occupies a band of adjacent depths. The lobula plate again illustrates the pattern, sitting almost entirely on levels 2 and 3 with a thin extension into level 4, which is what one would expect of a structure that receives from one stage and projects to the next rather than one whose level assignment is noise.

The third is the comparison that settles the original question. If the layering were an artefact of having been calibrated on super-classes, it should align with super-classes better than with anything else. It does the opposite. The weighted purity of the super-class is 0.413, below the 0.563 of the anatomical group, and the normalised mutual information tells the same story more sharply: 0.144 for the

super-class against 0.301 for the group, with the super-region falling between them at 0.190. Knowing a neuron’s anatomical group removes more than twice as much uncertainty about its level as knowing its functional super-class does. Table 4 collects the figures and Figure 11 displays them.

The reason is not mysterious. The decision tree was trained to separate super-classes, but it could only separate them as well as the underlying spectral axis allowed, and that axis is a statement about connectivity: a neuron sits early on it if it receives from few and projects to many, late if the reverse. Anatomical groups track connectivity closely, since a neuropil is defined by where its neurons make synapses, whereas a super-class such as *central* contains structures at every depth of the brain and cannot track it at all. The layering is therefore not merely transferable from super-classes to groups; groups are the categorisation on which it is sharpest.

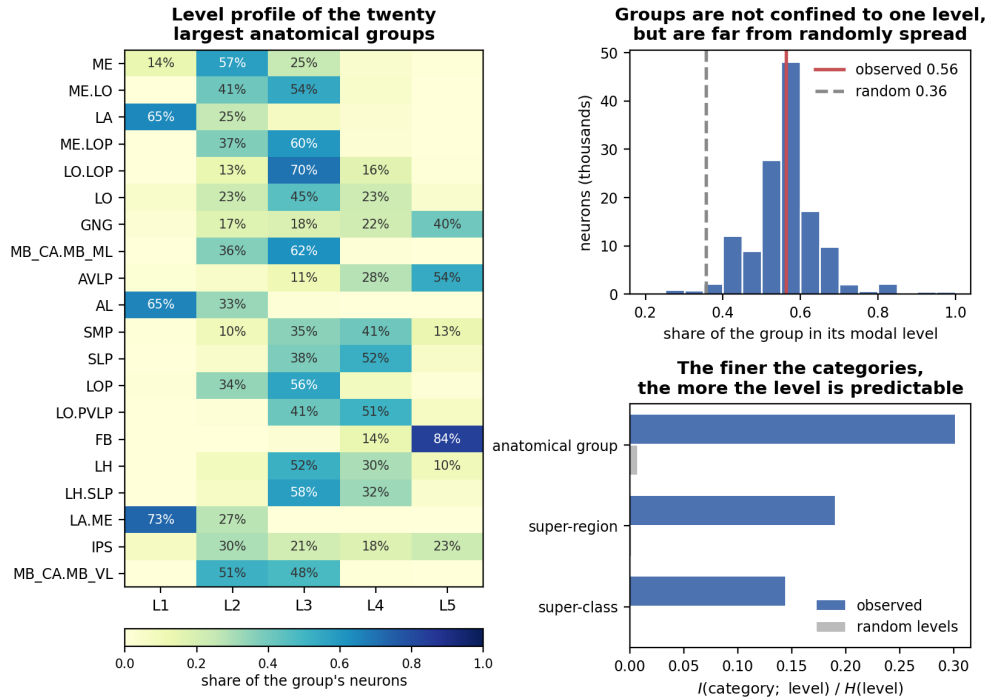


Figure 11: How anatomical categories distribute over the hierarchical levels. On the left, the level profile of the twenty largest anatomical groups, each row normalised to its own population: the mass of a group concentrates on adjacent levels rather than spreading uniformly, and the lobula plate is visible as a structure straddling levels 2 and 3. In the centre, the distribution of purity over all 628 groups weighted by population, against the value obtained when levels are assigned at random. On the right, the normalised mutual information between category and level at the three granularities: the finer the categorisation, the more the level is determined, which is the opposite of what one would expect if the layering were an artefact of the super-class labels it was calibrated on.

One caveat should be recorded. All three quantities are computed on the levels as given, and none of them validates the level construction itself; they establish only that the construction is at least as compatible with the anatomical group as with the categorisation used to build it.

Nor does the functional gradient of Section 4.1 supply the missing validation, for the reason given there: the level boundaries were chosen to separate super-classes, so finding that the levels separate super-classes is not an independent test. What that gradient does support is the *spectral embedding* underlying the levels, since four thresholds on a fixed axis could not have produced it had the axis not already carried the ordering. The discretisation itself remains a modelling choice inherited from earlier work, and no result in this thesis tests it.

4.3 The Macro-Scale Hourglass

Table 5 reports the hourglass score under both routing schemes and several bounds on path length, all at $\tau = 0.9$.

Table 5: Hourglass score under different routing schemes and path-length bounds, at $\tau = 0.9$. The last row is degenerate and is reported to show why the bound is mandatory.

Routing	Max hops	$S \rightarrow T$ paths	$ C(\tau) $	$ C_f(\tau) $	H
levels	4 (structural)	1,995,460	154	307	0.498
score	3	4,236,294	331	456	0.274
score	4	1.20×10^8	326	433	0.247
score	5	3.67×10^9	269	401	0.329
score	∞	1.17×10^{75}	1	10	(degenerate)

Under structural routing the score is 0.498, which is to say that the real core is half the size that the dependency structure alone would require. By the definition of Equation (9) the hourglass property is therefore present, and Figures 12 and 13 show that the finding does not hinge on the particular threshold chosen.

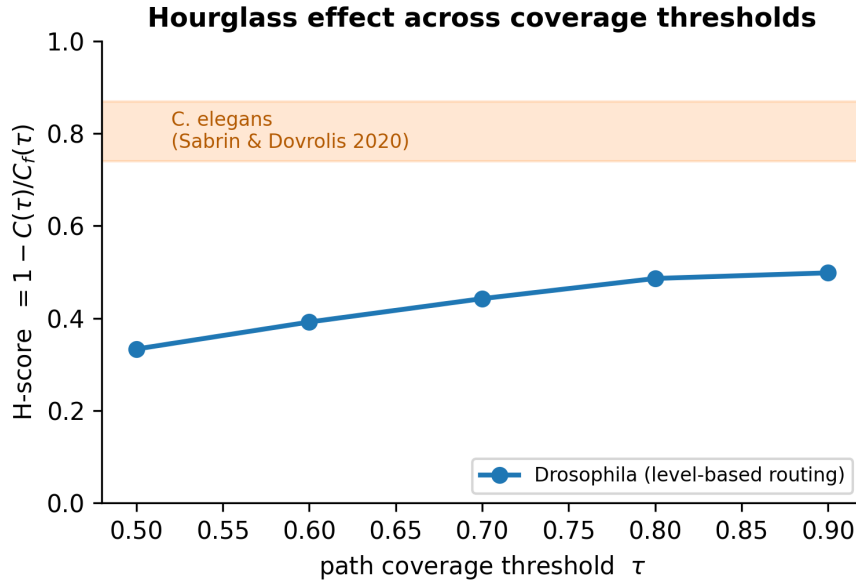


Figure 12: Hourglass score as a function of the coverage threshold τ , under level-based routing. The score is stable over a broad range of τ , which is what makes the single value reported at $\tau = 0.9$ meaningful rather than an artefact of one threshold.

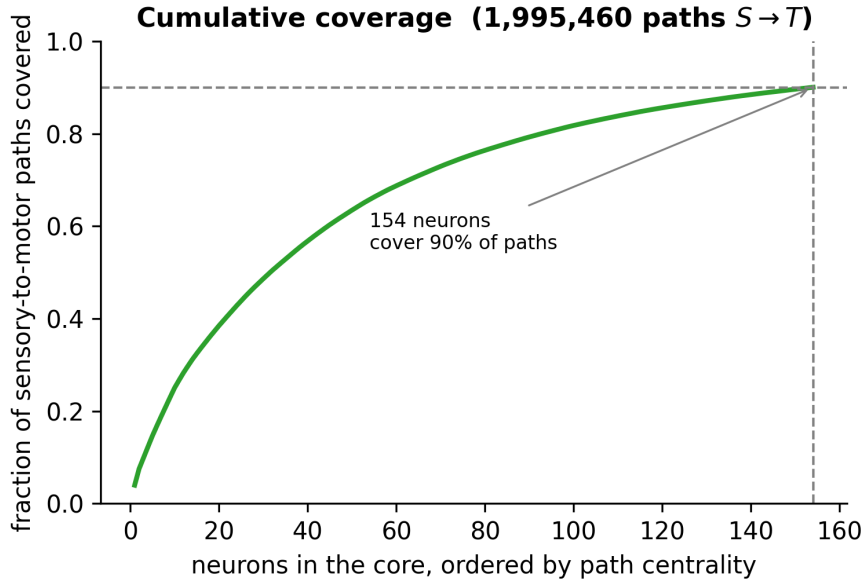


Figure 13: Cumulative fraction of sensory-to-motor paths covered as neurons are added to the core in decreasing order of path centrality. The sharp initial rise is the hourglass property in its rawest form: 154 neurons out of 134,181 carry 90% of the end-to-end flow.

The score is nevertheless routing-dependent, ranging from 0.247 to 0.498 across the schemes tested, and the honest way to report it is therefore the full table rather than any single figure drawn from it. The final row of the table shows what happens

when the bound on path length is removed altogether: the count of source-to-target paths reaches 1.17×10^{75} , a single neuron then covers 97.6% of them, and the τ -core collapses to one element. At that point the measure has lost all content, which is the concrete justification for the requirement stated in Section 3.2.

That the observed concentration is not an artefact of the degree sequence was checked directly against randomised networks, as Figure 14 shows.

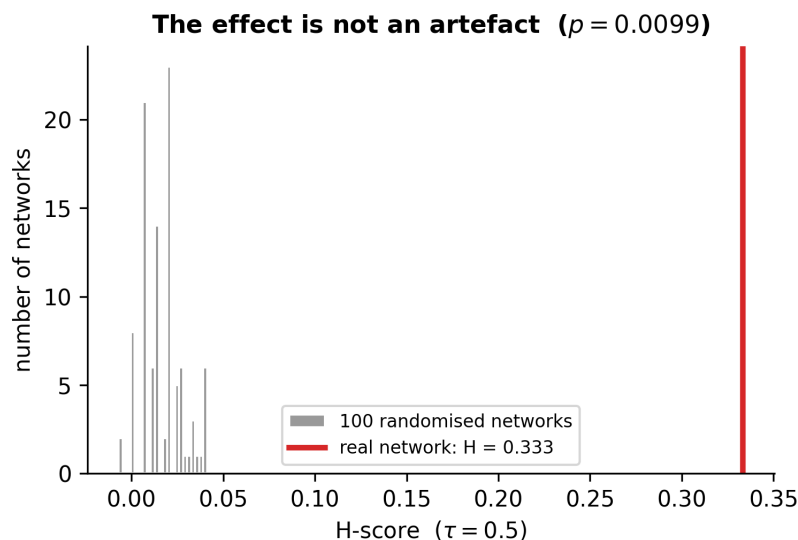


Figure 14: Distribution of the hourglass score over randomised networks, against the observed value. The observed score falls well outside the null distribution, establishing that the concentration of flow is not a by-product of the degree sequence alone.

4.3.1 Weaker Than in *C. elegans*, and Why That May Be Expected

The reference study of Sabrin et al. [11] reports hourglass scores between 0.74 and 0.87 for the *C. elegans* connectome. The values obtained here, between 0.25 and 0.50, are substantially lower. The *Drosophila* brain is therefore hierarchically organised, but it funnels its flow through a narrow waist considerably less than the nematode does, and the difference is large enough to require an explanation rather than a remark.

A plausible one, offered here as an argument to be weighed rather than as a demonstrated result, is that the two nervous systems are solving different problems. *C. elegans* routes essentially all of its sensory information through ten to fifteen command interneurons, an interneuron being a neuron that sits between others rather than receiving from a sense organ or driving a muscle; those few cells constitute a single shared bottleneck for the whole animal. The fly processes several sensory modalities in parallel, and each of them plausibly has its own bottleneck rather than sharing one. The composition of the τ -core supports this reading, since it

separates into distinct clusters, with AL structures serving olfaction, GNG and PRW serving taste and mechanosensation, and LH.SLP forming a third group, rather than forming a single shared core. Several parallel hourglasses will necessarily produce a lower global score than one deep hourglass, even when each individual pathway is strongly funnelled, because the union of several small cores is larger than any one of them.

4.4 Bow-Tie Motifs Across Level Windows

This section reports the exhaustive motif search. It covers the four level windows and what changes between them, the decomposition of the patterns by directional signature, and the distribution of the mass across chains of levels, which is what locates the waist within the hierarchy.

The organising finding is that the architecture is not uniform with depth. The shallow and the deep brain differ not in how many patterns they contain but in what kind, and the difference is large enough that reporting a single window would have given a misleading picture of the connectome.

4.4.1 The Four Windows

Table 6 summarises the exhaustive search over all four level windows, which to our knowledge has not previously been carried out on this connectome.

Table 6: Exhaustive bow-tie motif search across level windows, with $k_{\text{in}} = k_{\text{out}} = 2$, $\text{max_jump} = 1$ and $\text{min_count} = 10$.

Window	Neurons	Metanodes	Motifs	Runtime
$\{1, 2, 3\}$	104,915	879	31,184,800	14m39s
$\{2, 3, 4\}$	108,438	1155	119,026,375	57m43s
$\{3, 4, 5\}$	73,531	1220	433,546,629	3h05m51s
$\{1, 2, 3, 4, 5\}$	134,181	—	512,969,374	3h34m49s

The motif count grows steeply with depth, from thirty-one million through one hundred and nineteen million to four hundred and thirty-four million, while the number of neurons does not grow at all: the deepest window contains fewer neurons than the shallowest. Figure 15 makes the dissociation visible. Growth is therefore driven by connection density rather than by the size of the search space, exactly as the density measurements of Section 4.1 would predict.

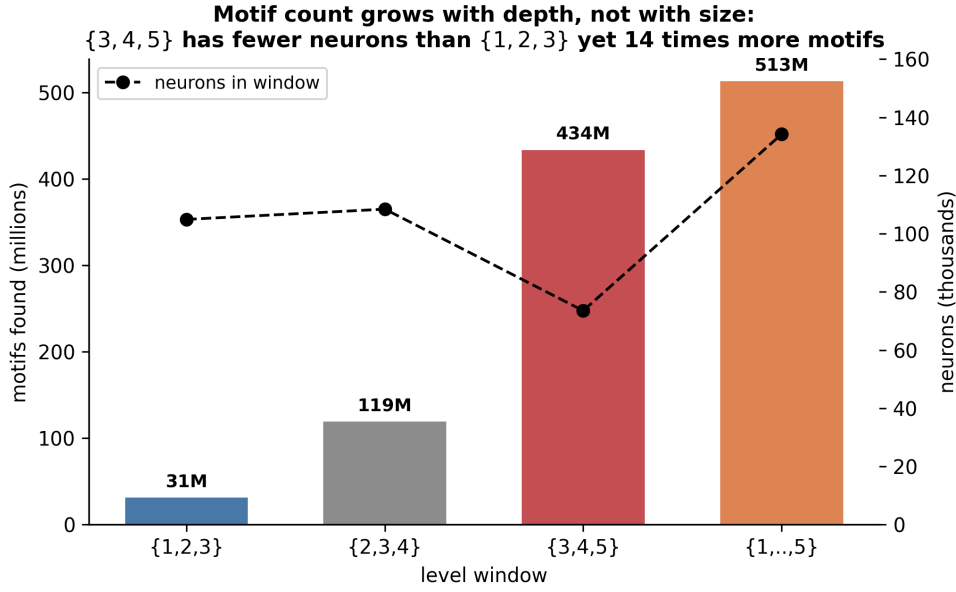


Figure 15: Number of bow-tie motifs found in each level window, shown as bars against the left axis, together with the number of neurons the window contains, shown as a dashed line against the right axis. The two do not track each other: window $\{3, 4, 5\}$ contains fewer neurons than $\{1, 2, 3\}$ yet yields fourteen times as many motifs.

The scale of the pruning described in Section 3.3 can be read off the shallow window, where the search examined 8,310,710,163 candidate combinations. Of these, 112,688,248 were excluded by the disjointness constraint and 8,125,079,064 were discarded because one of their flow factors was identically zero, leaving 72,942,851 raw motifs of which 31,184,800 passed the minimum count threshold. More than ninety-seven per cent of the nominal space is thus eliminated without any loss of exactness.

4.4.2 Directional Signatures

Section 2.3 set out why the motifs are classified by the signature of Equation (3) rather than by the agreement of all four edges at once. What that section could not show is how the thirteen classes hidden inside the coarse *mixed* label actually behave, and they turn out to differ considerably, as Figure 16 shows.

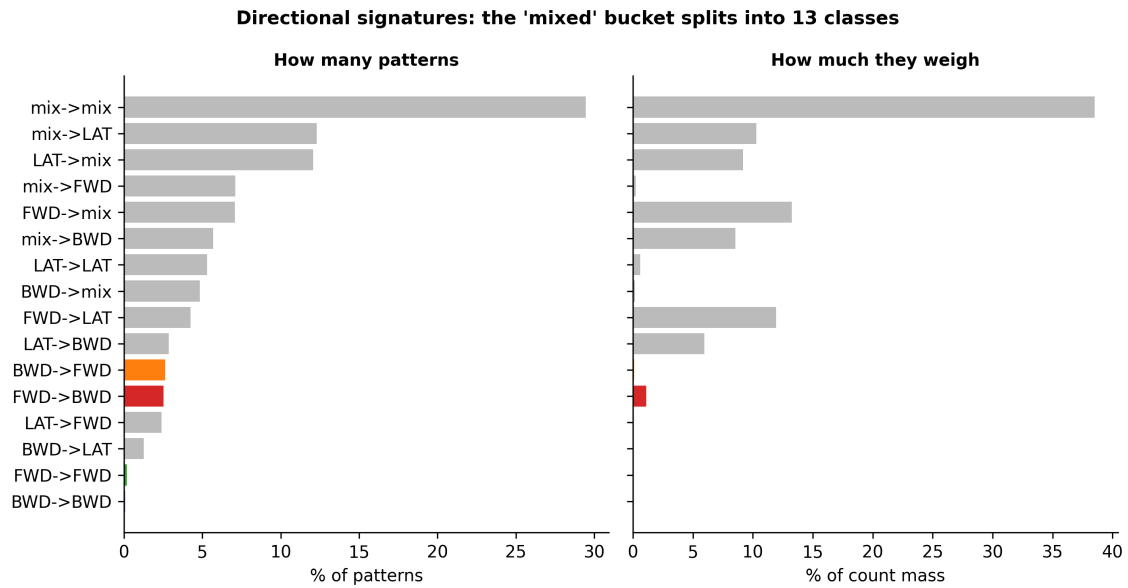


Figure 16: Decomposition of the sixteen directional signatures on window $\{1, 2, 3\}$, by number of patterns on the left and by share of total count mass on the right. The two panels disagree substantially: `mix->FWD` is frequent but carries almost no mass, while `FWD->mix` is the reverse. The pure signatures `FWD->FWD` and `BWD->BWD` are highlighted, and their rarity is what makes the coarse four-way classification uninformative.

The two panels of that figure disagree in an instructive way. A signature may account for many distinct patterns while contributing almost nothing to the total count, as `mix->FWD` does, or account for relatively few patterns each of which is realised enormously often, as `FWD->mix` does. Reporting only one of the two quantities would therefore give a misleading picture, and both are reported throughout.

Comparing the shallow and deep windows reveals that this composition is not a fixed property of the connectome but changes substantially with depth.

Table 7: Share of total motif mass by directional signature, shallow versus deep window. The upper block lists every signature in which at least one branch is entirely lateral, that is, in which *both* edges entering the waist, or *both* edges leaving it, run within a single level; the lower block gives the two largest signatures without such a branch. The seven signatures omitted from the table carry 10.2% of the shallow mass and 2.1% of the deep mass between them.

Signature	{1, 2, 3} (% mass)	{3, 4, 5} (% mass)
LAT->mix	9.2	42.7
mix->LAT	10.3	23.3
FWD->LAT	11.9	0.4
LAT->LAT	0.6	10.2
LAT->BWD	6.0	0.2
LAT->FWD	0.0	3.7
BWD->LAT	0.0	2.1
<i>all seven of the above</i>	<i>38.0</i>	82.6
mix->mix	38.5	13.9
FWD->mix	13.3	1.5

Signatures containing at least one purely lateral branch carry 38.0% of the mass in the shallow window and 82.6% in the deep one, while **mix->mix**, which dominates the shallow window at 38.5%, falls to 13.9%. What changes with depth is not only how much of the mass is lateral but which kind: in the shallow window the lateral branch is usually paired with a forward or backward one, as in **FWD->LAT** at 11.9%, whereas in the deep window it is almost always paired with a mixed branch or with another lateral one. Figure 17 displays the same comparison signature by signature, sorted by the magnitude of the shift. The deep brain is, in this precise sense, laterally dominated: its characteristic circuit is one in which information enters or leaves a waist by moving sideways within a processing stage rather than by advancing through the hierarchy.

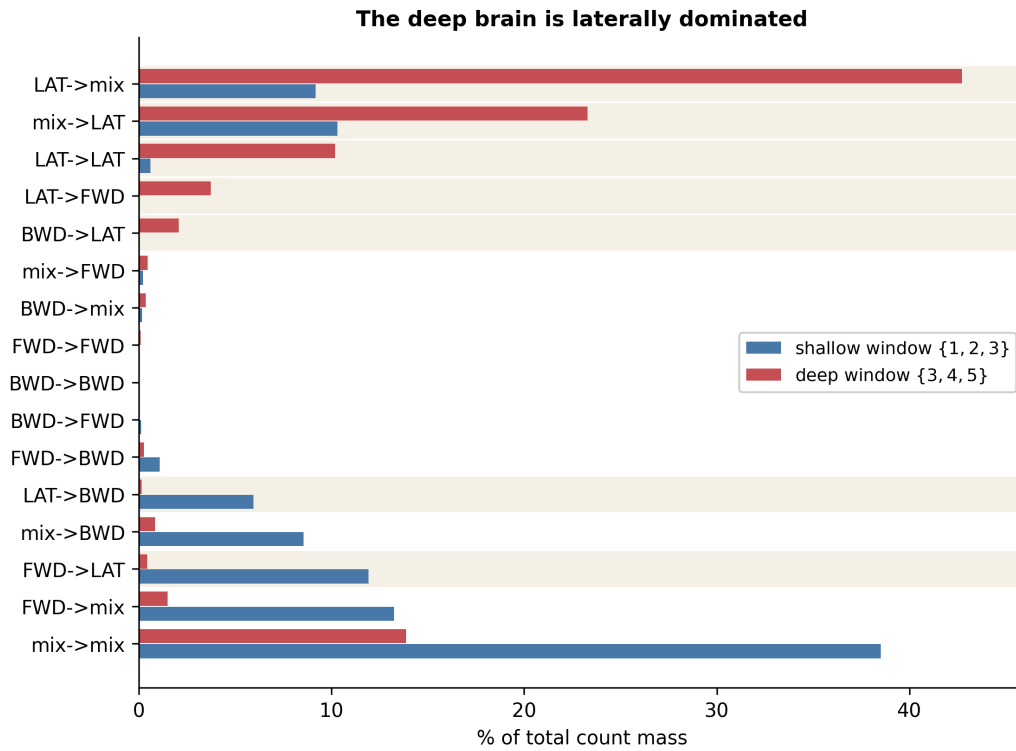


Figure 17: Share of total count mass by directional signature in the shallow and deep windows, sorted by the size of the shift between them. Shaded rows are signatures containing a purely lateral branch, and they gain almost exactly what the others lose. LAT->mix alone rises from 9.2% to 42.7%.

The highest-count individual patterns illustrate the same point concretely. Figure 18 shows the single most frequent bow-tie of the shallow window, and Figures 19 to 22 give representative examples of every signature class.

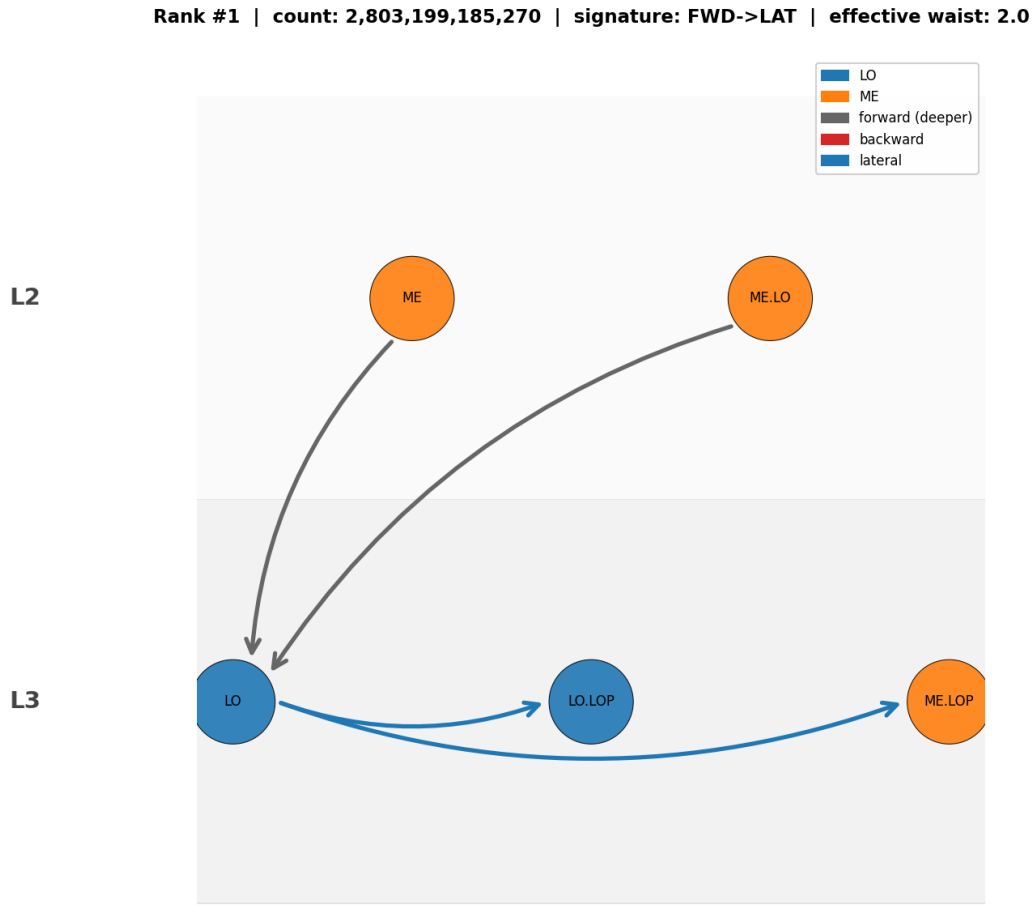


Figure 18: The single most frequent bow-tie motif of window $\{1, 2, 3\}$, with 2.80×10^{12} instances and signature FWD->LAT. The vertical axis is the hierarchical level, with depth increasing downwards and each level drawn as a shaded band, so that the direction of an edge is given by its slope: an edge that descends runs forward, one that rises runs backward, and a horizontal one is lateral. Two medulla groups at level 2 converge onto the lobula at level 3, which then distributes laterally within level 3. Grey edges are forward and blue edges lateral. The pattern is a compact illustration of the general finding: the dominant structures combine a forward entry with a lateral spread rather than being purely forward.

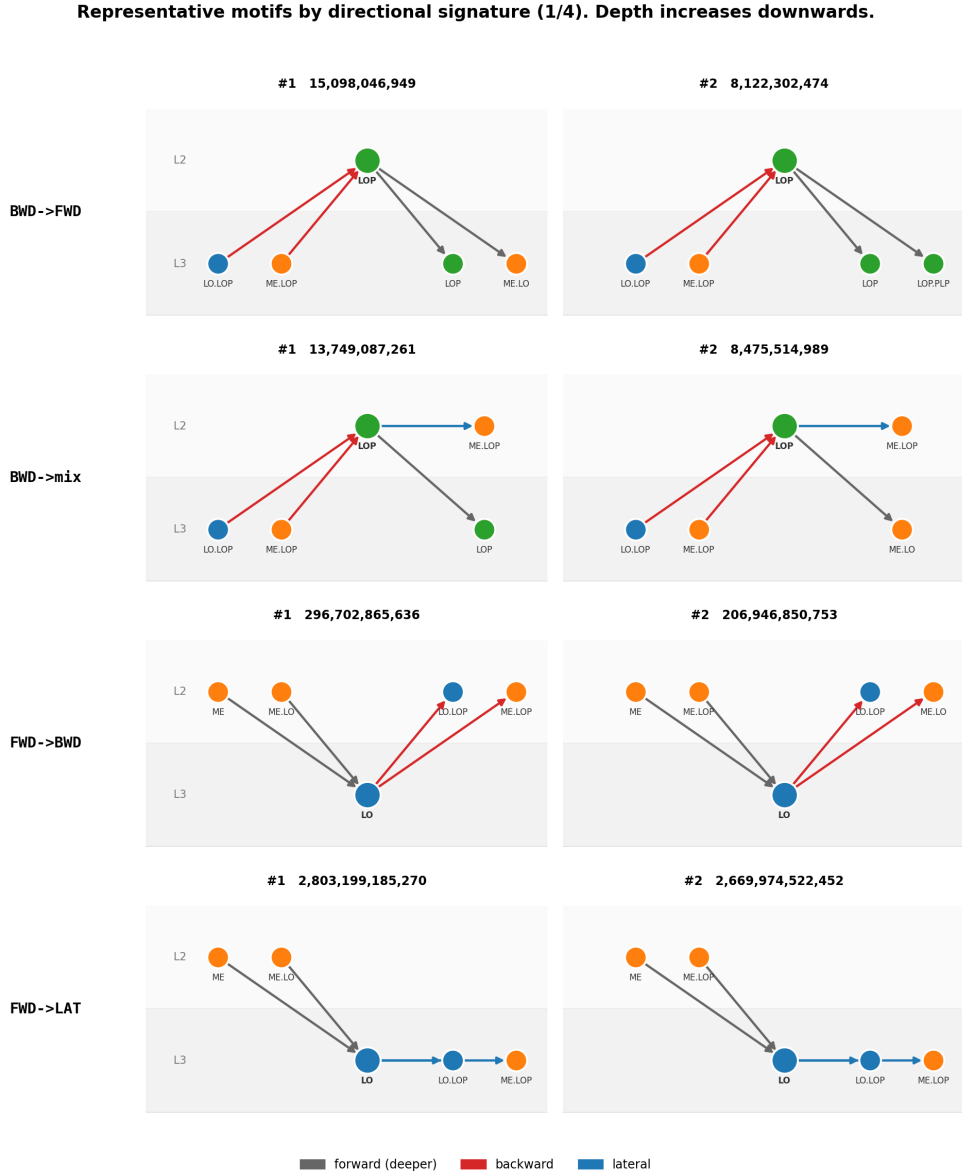


Figure 19: Representative bow-tie motifs for the first four directional signatures, window $\{1, 2, 3\}$. The horizontal axis carries the role a group plays in the motif, with the input groups on the left, the waist in the centre and the output groups on the right; the vertical axis carries the hierarchical level, drawn as shaded bands labelled at the left margin, with depth increasing downwards. The two axes together make the structure legible from the shape alone: an edge that descends is forward, one that rises is backward, and a horizontal one is lateral, so the signature can be read without consulting the node labels. Labels give the anatomical group, node colour encodes the macro-area, and edge colour repeats the direction, with grey for forward, red for backward and blue for lateral. The counts above each panel are numbers of neuron-level instances, and they differ between classes by nearly three orders of magnitude.

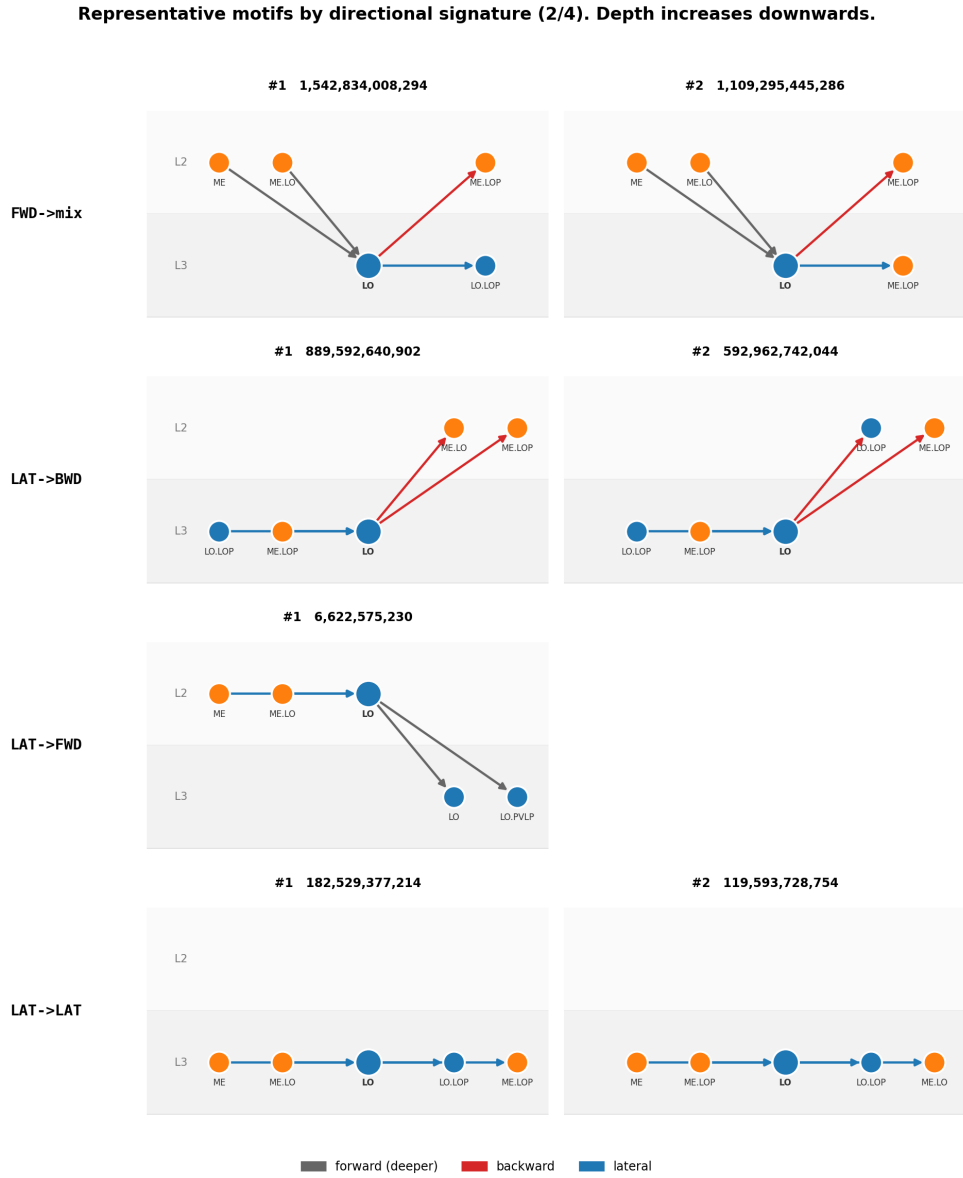


Figure 20: Representative motifs, signatures five to eight, with conventions as in Figure 19. The optic-lobe groups dominate the highest-count patterns throughout, consistent with the density reported in Section 4.1.



Figure 21: Representative motifs, signatures nine to twelve, with conventions as in Figure 19.

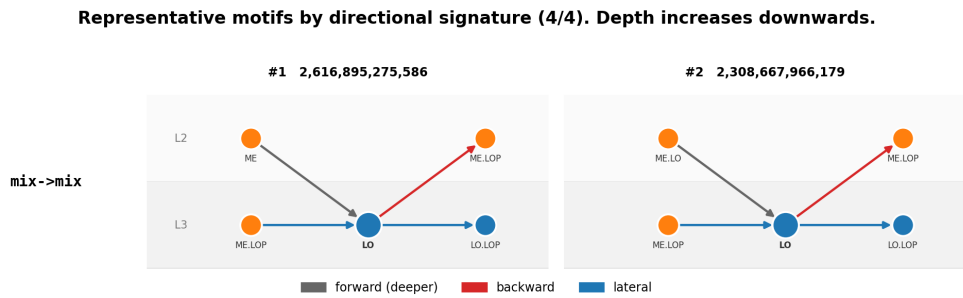


Figure 22: Representative motifs, the remaining signatures, with conventions as in Figure 19.

4.4.3 Level Chains

A complementary view attributes the mass of each pattern to the triple consisting of the level of its inputs, the level of its waist and the level of its outputs. This is coarser than the signature, since it records only which levels are involved rather than how the branches are oriented, but it answers a question the signature cannot: where in the hierarchy the bottlenecks actually sit.

In the deep window the dominant chains are L3 to L3 to L3 at 45.6% of the mass, L3 to L3 to L4 at 28.0%, L4 to L3 to L3 at 16.6% and L4 to L3 to L4 at 3.3%, as Figure 23 shows.

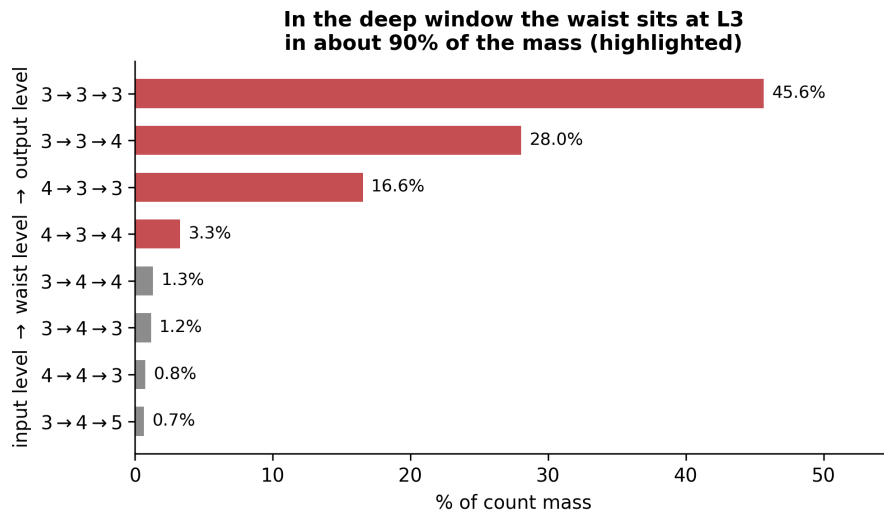


Figure 23: Distribution of motif mass across level chains in the deep window, where a chain records the input level, the waist level and the output level. Chains whose waist lies at level 3 are highlighted, and together they account for roughly 90% of the mass.

The waist sits at level 3 in roughly ninety per cent of the mass. That the bottleneck is in the middle of the hierarchy is the expected outcome rather than a surprising one, and it is worth saying why before drawing anything further from the table.

A bow-tie requires convergence on its way in and divergence on its way out. Neither is available at the ends of the hierarchy. Level 1 is dominated by sensory neurons, which receive almost nothing from inside the brain and so cannot be converged upon; level 5 is dominated by descending and motor neurons, which project out of the brain and so have little to diverge into within it. The middle is the only place where a neuron has both a substantial input from within the network and a substantial output back into it, and it is also, for the same reason, where the processing happens. Finding the waist there therefore does not so much reveal

something unexpected about the connectome as confirm that the measure is picking up processing structure rather than an artefact of how the levels were drawn. It is the result one would want to see before trusting anything else the search reports.

What is not automatic is which of the middle levels it turns out to be, and how sharply. Level 5 is the most internally dense level of the entire connectome, as Section 4.1 established, and yet it barely appears as a waist at all. High internal density combined with very few bottlenecks is a specific structural signature: it indicates connectivity that is recurrent and lateral rather than convergent and divergent. A region can be densely wired without any of its neurons functioning as an obligatory passage point, and level 5 appears to be exactly such a region. That distinction is invisible to density measurements taken on their own, and it is one of the clearer instances in this work of the two scales of analysis saying different things about the same tissue.

4.4.4 The Shallow Window, and a Direct Answer About L1

The same analysis on the shallow window gives a sharper picture still, in which four chains account for 99.2% of the entire count mass.

Table 8: Level chains in the shallow window: four chains carry almost all the mass, and all four have their waist at level 3.

Chain	Share of mass
L2 $\rightarrow$ L3 $\rightarrow$ L3	33.3%
L3 $\rightarrow$ L3 $\rightarrow$ L2	24.4%
L2 $\rightarrow$ L3 $\rightarrow$ L2	21.6%
L3 $\rightarrow$ L3 $\rightarrow$ L3	19.9%
total	99.2%

All four have their waist at level 3, the same concentration observed independently in the deep window and therefore not a peculiarity of either. Two windows sharing only one level agree on where the bottleneck sits, which is a stronger statement than either window makes alone.

The table also answers directly a question that motivated this part of the analysis, namely how many bow-tie patterns run from level 1 into level 2. The answer is that almost none do. Every chain involving level 1 together accounts for roughly 0.02% of the mass, despite level 1 containing 13,601 neurons. The explanation is functional rather than an artefact of the method: level 1 is 94% photoreceptors and

sensory neurons, which receive very little from within the brain and project forward. They are sources of flow rather than points of convergence, and a bow-tie requires convergence. A neuron population can therefore be large and still be a poor waist, which is a useful corrective to the intuition that the biggest structures must be the most important ones.

4.5 What the Choice of Granularity Recovers

Section 2.2.5 argued for grouping neurons by anatomical group rather than by functional super-class or by super-region. Since that choice conditions every result in this chapter, and since the coarsest of the three is the one adopted in the previous work of Section 2.5, the entire search was repeated at both of the other granularities on the shallow window, so that the comparison could be made on results rather than in the abstract. The three runs differ only in the map from neurons to categories; the window, the parameters, the engine and the pruning are identical.

Table 9: The same search on window $\{1, 2, 3\}$ under the three categorisations. Categories counts the values present in the window, and metagraph edges counts distinct ordered pairs of metanodes joined by at least one synapse within one level of each other.

	super-class	super-region	anatomical group
categories in window	9	76	477
metanodes	24	188	879
metagraph edges	293	6,295	24,914
median neurons / metanode	511	11	3
distinct waists	20	157	660
motifs found	29,091	4,923,197	31,184,800
runtime	6.7 s	2m31s	14m39s

The first thing to report is what the coarse analysis does *not* do, because it contradicts an argument that might be expected. One could suppose that super-class granularity is unusable because its metanodes are too large to be waists: the largest, *optic* at level 2, contains 35,584 neurons, a quarter of the network. That supposition is wrong. The waist carrying the most mass in the coarse run is *visual centrifugal* at level 2, a metanode of only 311 neurons with an effective size of 3.9. Narrow waists emerge perfectly well at coarse granularity, because what produces a high count is convergence and divergence structure, not population. The coarse analysis is

also very cheap, finishing in under seven seconds and yielding twenty-nine thousand patterns rather than thirty-one million, which makes it far easier to inspect by hand.

What the coarse analysis loses is anatomical resolution, and the clearest way to see it is to put the leading pattern of the three runs side by side. At super-class granularity the most frequent bow-tie reads

$$[\text{central L1, central L2}] \rightarrow \text{visual centrifugal L2} \rightarrow [\text{optic L1, optic L2}],$$

which says that central neurons converge on visual centrifugal neurons that project to optic ones. At super-region granularity it reads

$$[\text{AL_L1, AL_L2}] \rightarrow \text{AL.MB_L2} \rightarrow [\text{MB_L2, MB_L3}],$$

which is already a recognisable circuit: two antennal lobe populations converge on neurons spanning the antennal lobe and the mushroom body, which then project into the mushroom body. That is the olfactory pathway, and it is the pathway on which several of the results of this chapter turn. At group granularity the leading pattern reads

$$[\text{ME_L2, ME.LO_L2}] \rightarrow \text{LO_L3} \rightarrow [\text{LO.LOP_L3, ME.LOP_L3}],$$

which names a specific circuit within the optic lobe: two medulla populations converge on the lobula, which distributes within its own level to the lobula plate. All three statements are true of the same connectome; they differ in how much they commit to.

The gap between the first and the other two is not a matter of degree. The category *central* contains 521 distinct anatomical groups, and the antennal lobe projection neurons, the Kenyon cells, the mushroom body output neurons and the lateral horn all fall within it without exception. The olfactory sequence just described therefore has every one of its stages inside a single super-class node, and this is why the coarse run does not merely rank it lower but cannot express it at all. The same holds for the sequence reported in Section 4.10, in which projection neurons broadcast onto Kenyon cells that then feed integrative output neurons; for the compression results of Section 4.6, which identify particular neuropils as bottlenecks; and for the recurrence result of Section 4.6.1, which separates re-entry into the lobula from relay through the lobula plate. None of these survives the collapse of the central brain into one node.

The super-region granularity is the interesting middle case, because it recovers

most of what matters at a fraction of the cost. It is fine enough to name the olfactory pathway, to separate the optic lobe from the central complex and to distinguish the mushroom body from the lateral horn, and it does so with 188 metanodes rather than 879 and in a sixth of the runtime. What it cannot do is separate structures within a super-region: the calyx, the pedunculus and the two lobes of the mushroom body are one node, so a claim about where in the mushroom body a bottleneck sits is unavailable, and the medulla, lobula and lobula plate are likewise merged, which is fatal to the recurrence analysis of Section 4.6.1. It is the right granularity for an overview and the wrong one for the questions this thesis asks, but it is a genuinely useful third point of reference, and its existence means the choice made here is not a binary one between an unusable coarse view and an over-specific fine one.

4.5.1 What the Abstraction Actually Buys

A fair objection to the fine granularity is that it weakens the abstraction as a device for reducing complexity. If there are six hundred possible categories per level, the metagraph starts to look less like a summary of the connectome and more like a relabelling of it, and the reduction that justified moving from neurons to metanodes in Section 2.2.5 is correspondingly smaller. The objection is sound in direction, and the honest answer is to quantify it rather than to deflect it.

On the shallow window the connectome contributes 104,915 neurons and 1,296,899 distinct neuron-to-neuron connections within one level of each other. The three abstractions reduce the nodes by factors of 4,371, 558 and 119 respectively, and the edges by factors of 4,426, 206 and 52. The finest choice therefore still removes two orders of magnitude from both, which is not nothing, but it removes a great deal less than the coarsest one, and the number of metanodes it leaves, 879, is of the same order as the number of anatomical groups themselves. A reader who expected the abstraction to leave a graph small enough to be inspected whole will not find one.

Two things should be said in response. The first is that the reduction matters less than it appears, because the search cost is not driven by the number of metanodes but by the number of combinations of them around each waist, and the pruning of Section 3.3.2 removes most of those before they are ever formed. The shallow window runs in fifteen minutes at 879 metanodes, so the abstraction is doing enough work to make an exhaustive count feasible, which was its purpose.

The second, and the more important, is that reduction was never the only objective. The abstraction has two jobs, and they pull against each other: it must make the count tractable, and it must leave the result interpretable. A metagraph of twenty-four nodes is maximally reduced and tells us that *central* neurons feed

visual centrifugal ones, which is not a circuit. A metagraph of 879 nodes is less reduced and tells us that the medulla feeds the lobula, which is. The right measure of an abstraction is not how much it compresses but how much of the compression it achieves without destroying the distinctions the question depends on, and by that measure the evidence in this section favours the finer choice, with the super-region available as an intermediate whenever a coarser summary is wanted. The three granularities are complementary rather than competing: reporting all three costs less than three minutes of computation, and it replaces an argument about which one is correct with a measurement of what each one recovers.

4.6 Waist Compression

The nominal compression metric fails in exactly the way anticipated in Section 3.3.3. Its ranking is headed by `L0.PVLP_L2`, which attains a nominal compression of 2431 with only *two* active neurons out of twenty-one, while the biologically prominent `AL.MB_CA_L2` scores merely 40. The first of these is not a bottleneck in any meaningful sense; it is a pair of neurons in a group that happens to be large.

Under the realised metric of Equation (10) the ranking becomes interpretable. On the shallow window `AL.MB_CA_L2` has 144 neurons in its group, of which 118 are active, an effective count of 4.8, and 2694 peripheral neurons connected through it, giving a realised compression of 561.8. The metric now reflects the structure one wants it to reflect, namely how much of the periphery is funnelled through how many genuinely participating neurons.

On the deep window the analysis retains 19,416 patterns, of which 794 have an effective waist of at least five neurons, and the internal consistency check described in Section 3.6 reproduced all 19,416 counts exactly. The leading structure there is `ME_L4`, with a realised compression of 891.8 over an effective waist of 5.7 neurons, followed by several variants of `ME.L0_L3`: the optic lobe dominates compression in the deep brain as it dominates the motif counts. Figure 24 plots the whole distribution.

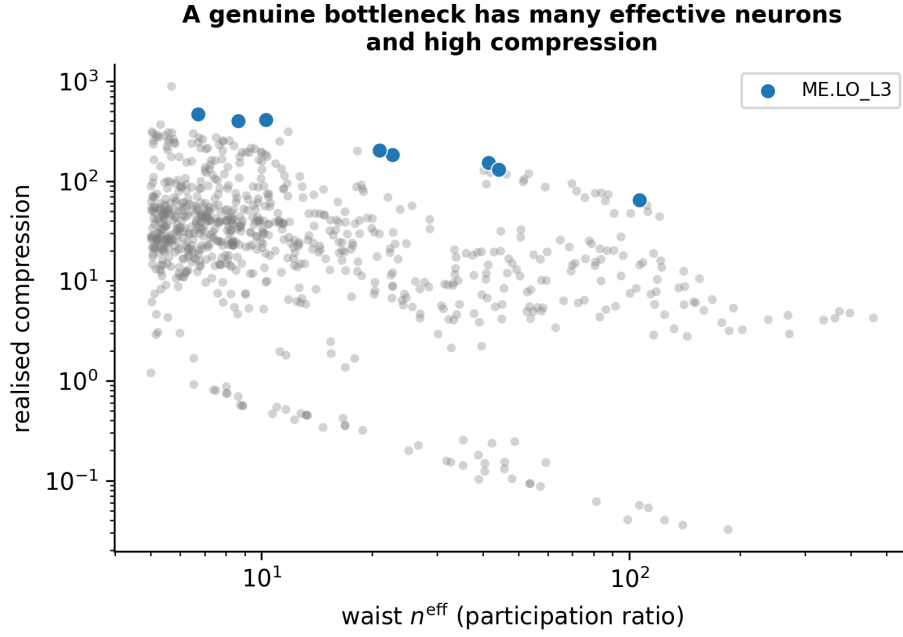


Figure 24: Realised compression against the effective number of waist neurons for window $\{3, 4, 5\}$, with the ME.LO_L3 family highlighted. Points at the far left achieve high compression through a single active neuron and are precisely the artefacts that the participation ratio is designed to expose. A genuine bottleneck lies towards the upper right, having both many effective neurons and high compression.

4.6.1 Two Anatomically Distinct Forms of Recurrence

Restricting attention to the two recurrent signatures reveals something that the aggregate figures conceal, namely that the two do not live in the same place.

The re-entrant form, FWD→BWD, comprises 786,216 patterns, which is 2.52% of the total, carrying a mass of 1.12×10^{12} or 1.10%. Its waist is the lobula, LO_L3, in 88.5% of that mass, and its leading pattern has two medulla groups converging onto the lobula which then projects onto two lobula plate groups. The relay form, BWD→FWD, comprises a very similar number of patterns, 822,466, but carries nine times less mass, and it sits on a different structure altogether: the lobula plate, LOP_L2, in 93.2% of its mass. The purely backward form, BWD→BWD, is negligible in mass and also lives on LOP_L2.

The two forms of recurrence are therefore anatomically separated rather than being two descriptions of one phenomenon. Re-entry into the lobula and relay through the lobula plate are distinct circuits, and both belong to the optic lobe, which dominates the recurrent structure of this window as it dominates much else.

This dominance survives the correction of the compression metric, but it must be qualified in one respect. Under realised compression the top of the ranking remains

optic, with ME.L0_L3 at 2173, ME.L0_L2 at 796 and ME_L3 at 690, but now with substantial counts, between 4×10^5 and 1.4×10^7 , and with genuine participation rather than with two active neurons out of twenty-one. The dominance is thus a property of the circuitry rather than an artefact of group size, which is a stronger claim than the nominal metric could have supported.

4.7 Statistical Significance

Of the three hundred and twenty pre-registered patterns, all three hundred and twenty had their observed counts reproduced exactly from the graph before any testing began. After Benjamini–Hochberg correction at $q < 0.05$, significance was obtained for 319 of 320 patterns under model A and for 310 of 320 under both models B and C.

That the strictest model yields nearly the same number of significant patterns as the weakest is the informative outcome. Model C preserves both the block structure and the per-block degree sequences and randomises only the alignment between them, so a pattern that remains significant under it cannot be explained by degree sequences, nor by regional connection densities, nor by which particular waist neurons happen to carry which degrees. The convergence–divergence structure of these bow-ties is therefore not accounted for by any of the three confounds tested.

4.7.1 The Ranking Between Signatures Inverts Between Null Models

The counts of significant patterns are similar across the three models, but the *ordering* of signatures by degree of enrichment is not merely different between them: it inverts.

Table 10: Median ratio of real to randomised count, by signature, under two complementary null models. The extremes swap places.

Signature	Null A (degrees kept)	Null B/C (anatomy kept)
FWD→FWD	2.21×10^5 (<i>highest</i>)	8.4–11.4 (<i>lowest</i>)
mix→FWD	2.06×10^3	4.7×10^4
BWD→BWD	1.98×10^3	9.2×10^3
LAT→LAT	1.84×10^3	6.4×10^5

The signature FWD→FWD is the most enriched of all when degree sequences are preserved and anatomy destroyed, and the *least* enriched when anatomy is preserved and the degree alignment destroyed. The signature LAT→LAT moves in the opposite direction by more than two orders of magnitude.

The interpretation is straightforward once stated. Purely forward bow-ties are rare given the degree sequence alone, so a null model that ignores anatomy finds them strikingly over-represented in the real network; but once the regional block structure is preserved, they turn out to be close to what that structure already implies, because the anatomy itself runs broadly forward. Lateral bow-ties behave in exactly the opposite way: unremarkable given the degrees, strongly over-represented given the anatomy.

This is the strongest argument this thesis can offer for using several null models rather than one. A single null would not merely have given a weaker answer; depending on which one had been chosen, it would have supported the opposite ranking of signatures, with nothing internal to that analysis to indicate that anything was amiss.

4.8 Micro–Macro Cross-Validation

Comparing the two scales on the six hundred and sixty groups of the shallow window gives a Spearman rank correlation of $\rho = +0.121$. The Spearman coefficient compares the two orderings rather than the two sets of values, which is what is wanted here, since the motif mass and the path coverage are measured on scales that are not comparable and the question is only whether a group ranking high on one tends to rank high on the other. Its value is with $p = 0.0018$ between the motif mass of a group acting as a waist and its coverage in the τ -core. A rank correlation was used rather than a linear one because both quantities span several orders of magnitude and neither is normally distributed; what it measures is whether groups that rank highly on one scale tend to rank highly on the other. The value is small but reliably different from zero, which means there is a real signal and that it is far from a one-to-one relationship.

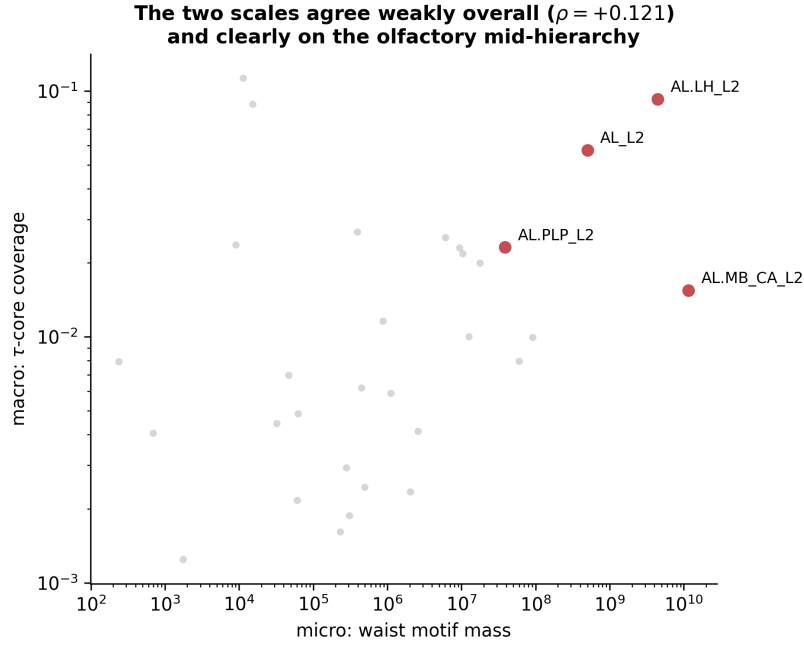


Figure 25: Waist motif mass at the local scale against τ -core coverage at the global scale, one point per group and level, with both axes logarithmic. The overall association is positive but weak. Highlighted are the mid-hierarchy groups of the olfactory pathway, on which the two scales agree closely.

Where the two scales agree, they agree on the mid-hierarchy waists of the olfactory pathway, as Table 11 shows.

Table 11: Groups where the two scales agree: mid-hierarchy waists of the olfactory pathway, with ranks out of six hundred and sixty groups.

Group	Micro rank (motif mass)	Macro rank (coverage)
AL.MB_CA_L2	8	12
AL.LH_L2	10	2
AL_L2	20	4
AL.PLP_L2	50	8

The overlap between the two top-twenty lists is three groups out of twenty. Two lists of twenty drawn at random from six hundred and sixty would be expected to share 0.61 groups, a figure given by the hypergeometric distribution, which describes how many elements two subsets of fixed size share when drawn without replacement from the same population; observing three gives $p = 0.0195$. That agreement is modest in absolute terms but is concentrated on structures of known functional importance, which makes it more meaningful than the raw number suggests.

The scales disagree on the input-side groups, and the disagreement is worth examining rather than dismissing. Groups such as GNG and PRW at level 1 are central in the global flow, appearing in the τ -core, while ranking very low as local waists. There is no contradiction here once the two measurements are read carefully: being an obligatory passage point on the way from sensation to action is a different property from being a point at which many streams converge and then diverge locally. A relay through which everything must pass will score highly on path centrality and poorly on bow-tie counts, and the fly’s early sensory structures are precisely such relays.

On the deep window the global correlation is comparable, at $\rho = +0.103$ with $p = 0.001$, but the overlap of the top-twenty lists falls to one group out of twenty and is no longer significant, with $p = 0.33$. This is the expected outcome rather than a failure of replication, since the τ -core is dominated by olfactory and gustatory pathways which reside at levels one to three and are therefore largely outside the deep window.

4.9 Neurotransmitters: A Hypothesis and Its Scope

The waists identified so far are structural objects, and a natural question is whether they are also chemically distinct: whether the waists of backward circuits, in particular, are more inhibitory than those of forward ones. The hypothesis is plausible, since inhibition is commonly associated with feedback control, and the data needed to test it are available, because FlyWire predicts a neurotransmitter for every neuron, under the classification set out in Section 2.2.2.

This section reports what came of it. Part of the answer is robust and is stated first; the central comparison holds on one window and not on the others, and the reason for the discrepancy is identified rather than left open; and what survives is a narrower claim about one brain region, together with the procedure that separated it from the broader one.

4.9.1 What Holds

Before turning to what does not hold, it is worth stating what does. The composition of excitatory and inhibitory neurons by anatomical group is robust across every window examined: AL.MB_CA and the mushroom body are essentially entirely excitatory, while the optic lobe and the lateral horn are the most inhibitory structures in the brain. This result depends only on counting neurons by their predicted neurotransmitter within each group, and not on any comparison between classes of motif, which is why it survives the difficulties described below.

Figure 26 shows the related and also robust observation that forward edges are more excitatory than backward ones. That difference, however, is almost entirely attributable to the medulla, whose edges dominate the forward class, and it is a statement about edges rather than about waists.

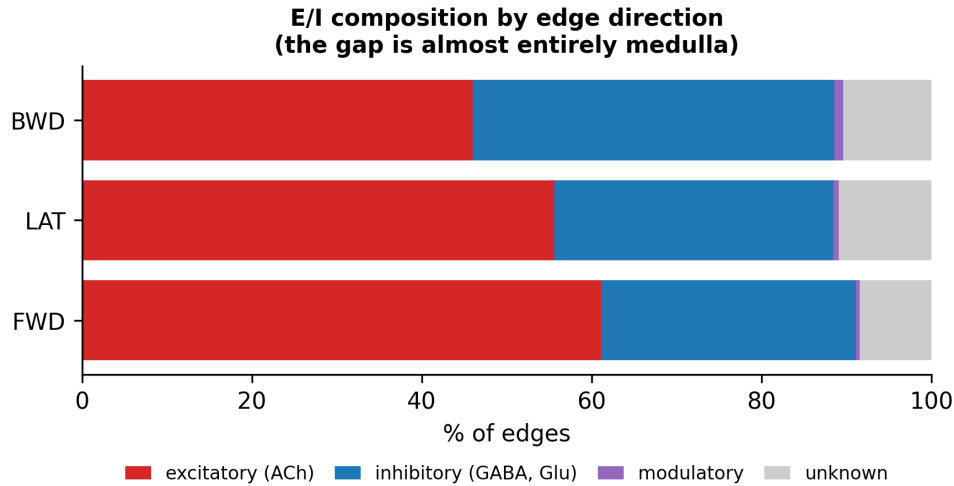


Figure 26: Excitatory and inhibitory composition of edges by direction. Forward edges are more excitatory than backward ones, but the difference is almost entirely attributable to the medulla. This figure concerns edges; the waist-level comparison that follows, which is the hypothesis actually under test, does not survive replication.

4.9.2 What Does Not

The hypothesis under test was that the waists of backward bow-ties differ neurochemically from those of forward bow-ties. The natural way to test it is to compare, for each waist group that hosts motifs of both kinds, the excitatory fraction of that group as it appears in the two classes; working at parity of waist in this way controls for the regional composition that Section 1.2 identified as the principal confound.

Because each waist contributes one value to each class, the two samples are paired rather than independent, and the comparison is made with the Wilcoxon signed-rank test: the differences within each pair are ranked by magnitude and the ranks of the positive and negative differences are compared. The test is used rather than a paired t -test because the excitatory fractions are proportions bounded in $[0, 1]$ and far from normally distributed, with many waists entirely excitatory or entirely inhibitory. Every p -value reported in this section comes from that test.

On the shallow window the effect appears clear, with a gap of $+0.122$ at $p = 0.0034$. Extending the same comparison to all four windows produces Table 12 and Figure 27.

Table 12: Excitatory fraction of the waist, *pure_BWD* against *pure_FWD*, compared at parity of waist across all four level windows.

Window	Shared waists	<i>pure_BWD</i>	<i>pure_FWD</i>	Gap at parity	<i>p</i>
{1, 2, 3}	86	0.387	0.508	+0.122	0.0034
{3, 4, 5}	221	0.501	0.547	+0.046	0.0098
{2, 3, 4}	185	0.506	0.449	−0.057	0.0296
{1, 2, 3, 4, 5}	492	0.483	0.504	+0.021	0.098

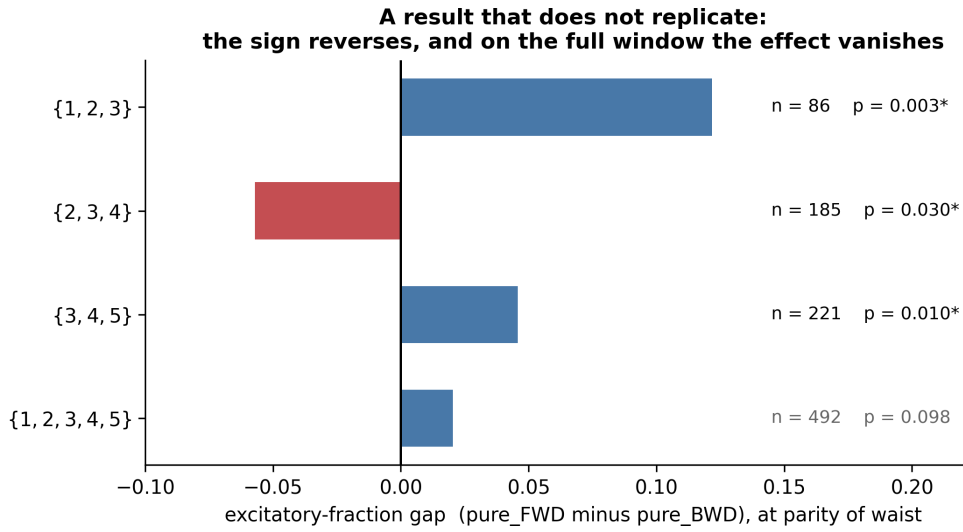


Figure 27: Excitatory fraction gap between *pure_FWD* and *pure_BWD* waists, measured at parity of waist, across all four level windows, with asterisks marking $p < 0.05$. Window {2, 3, 4} falls on the opposite side of zero from the others, and the full window, which has the most shared waists and subsumes the rest, does not reach significance.

Three observations together dismantle the conclusion. On window {2, 3, 4} the sign of the effect reverses, with a p -value that would conventionally be called significant: there, backward waists are *more* excitatory rather than less. On the full window, which has the largest number of shared waists and which subsumes all the others, the effect does not reach significance at all. And the raw gaps on those two larger windows, before any control is applied, are essentially zero, at +0.0003 and +0.0021 respectively, against +0.726 on the shallow window; there is consequently not even an effect present to be explained.

4.9.3 Why the First Window Looked Convincing

The explanation lies in where the two classes of motif happen to sit. On the shallow window the *pure_FWD* motifs concentrate in `AL.MB_CA`, which is essentially fully excitatory, while the *pure_BWD* motifs concentrate in the optic lobe, which is inhibitory. The apparent difference between motif classes is thus a difference between brain regions wearing the costume of a difference between circuit types.

The control at parity of waist did substantial work: it removed 83% of the raw gap on that window, taking it from +0.726 down to +0.122. What it did not do was remove all of it, and the residue looked like a genuine effect. In hindsight the residue was the remaining part of the same regional artefact, not an independent phenomenon, and the way this became visible was replication rather than any refinement of the control itself.

It is also worth noting the multiple-testing situation. Across four windows and six pairwise comparisons there are twenty-four tests in total, so roughly one false positive is expected at a nominal $\alpha = 0.05$. None of the four *p*-values in Table 12 survives a Bonferroni correction within its own window, except the one obtained on the shallow window. The Bonferroni correction is the most conservative way of accounting for multiple tests: it divides the significance threshold by the number of tests performed, so with six comparisons per window the level required drops from 0.05 to 0.0083.

4.9.4 What the Analysis Establishes

Three things come out of this, and they should be kept apart.

The first is a claim that holds. Within the medulla, forward connections are excitatory in 68.8% of cases and backward ones in only 34.8%. This is measured at parity of region, so it is not a composition effect, and it is biologically unsurprising: the medulla contains the motion-detection circuitry, in which lateral and feedback inhibition are the expected mechanism. This is the version of the hypothesis that the data support.

The second is a claim that does not hold, at least not on this evidence: that the same relation is a general property of the connectome. It is not supported outside the optic lobe, and the analysis says why rather than merely reporting that it failed. The two classes of motif are not spread evenly across the brain, so a comparison between them is partly a comparison between the regions they happen to occupy, and once that is controlled for most of the effect is gone.

The third is a question left open. Whether some weaker version of the relation holds in the central brain, where the two classes of motif are too unevenly distributed

for the present design to test it, is not settled here. Answering it needs a different design rather than more data: a set of waists selected in advance to carry both directions in comparable numbers, which this analysis, working from an exhaustive search rather than a targeted one, could not assemble. Section 5.3 returns to this.

The procedure itself is the part that transfers. It has three steps: control at parity of waist, decompose the raw gap to see how much of it the control removes, and replicate across independent subsets of the data. The last step was the decisive one, and its lesson is worth stating plainly. Two concordant windows are not a replication but two draws from overlapping data; it was only the third and fourth windows that made the scope of the claim visible.

4.10 Integrative, Broadcast and Mixed Networks

Everything so far has been counted with two inputs and two outputs, which makes every pattern symmetric by construction. That symmetry is a restriction on what can be seen, not a property of the connectome, and lifting it turns out to reveal something the symmetric search cannot: not just *where* the bottlenecks are, but *what kind of operation* each one performs.

Allowing k_{in} and k_{out} to vary independently distinguishes three regimes. A circuit with many inputs and one output is *integrative*: it gathers information from several sources and compresses it into one channel, which is what a decision or a classification looks like structurally. A circuit with one input and many outputs is a *broadcast*: it takes one signal and distributes it widely, which is what an expansion into a higher-dimensional representation looks like. Balanced configurations are mixed. The distinction is captured by an *integration index*, negative for broadcast and positive for integration, and the question this section asks is whether the index, computed from wiring alone, recovers operations that are known independently from physiology.

Enumeration is again avoided, this time by a different route. The sum over all k -combinations of a product of flow vectors is the elementary symmetric polynomial e_k of those vectors, which can be evaluated in $O(p \cdot k)$ time per neuron for p available groups without ever forming a single combination explicitly. The complete sweep for k from two to five on the shallow window takes ninety-four seconds, where explicit enumeration would have required forming 2.9 million combinations for a single waist at $k_{\text{in}} = 3$.

One methodological caution applies to the resulting index, and it is instructive because the artefact is easy to miss. The raw integration index is confounded with the number of groups available to a waist, since e_k over p groups grows combinatorially

with p : its Spearman correlation with $\log(p/q)$, where q is the number of output groups, is $+0.535$ at $p = 3.6 \times 10^{-38}$. A waist with many neighbours would therefore appear integrative purely on that account. Normalising by the number of available combinations reduces the correlation to -0.023 at $p = 0.61$, which eliminates the artefact, and only the normalised index is interpreted here.

4.10.1 The Sweep Recovers the Mushroom Body Architecture

The window contains eight hundred and fifty-three metanodes, of which seven hundred and five carry a total count of at least 10^3 and are classified; the rest are too sparsely populated for the index to mean anything. Of those seven hundred and five, two hundred and forty-eight come out integrative, three hundred and seventeen mixed and one hundred and forty broadcast. A tally of that kind is not by itself informative: any classification of five hundred objects into three bins will produce numbers, and nothing in them says whether the bins mean anything.

What makes the classification testable is that there exists a circuit whose answer is known in advance. The olfactory pathway of the mushroom body is among the best characterised in any brain, and its computational organisation has been established physiologically rather than structurally: a compact odour code carried by a few dozen projection neuron types is expanded onto a much larger population of Kenyon cells, and that expanded representation is then read out by a small number of output neurons whose activity drives behaviour. Expansion followed by compression is exactly the sequence broadcast $\rightarrow$ mixed $\rightarrow$ integrative would describe, so the pathway constitutes a prediction the index either meets or fails.

Table 13 therefore reports the five metanodes lying along that pathway rather than the highest-scoring metanodes overall. The selection is made anatomically and before looking at the index: they are the waists corresponding to the antennal-lobe projection neurons, to the parallel projection into the lateral horn, to the Kenyon cells, and to the two mushroom body output populations. Reporting the extremes of the ranking instead would have shown only that the index has extremes.

Table 13: Normalised integration index along the olfactory pathway. Negative values indicate broadcast, positive values integration.

Waist	in / out	Index	Regime
AL.MB_CA_L2 (projection neurons)	46 / 99	−0.895	broadcast
AL.LH_L2	100 / 192	−0.712	broadcast
MB_CA.MB_ML_L3 (Kenyon cells)	58 / 53	+0.061	mixed
MB_ML.CRE_L2 (MB output)	45 / 47	+0.997	integrative
MB_PED.CRE_L2	28 / 29	+1.000	integrative

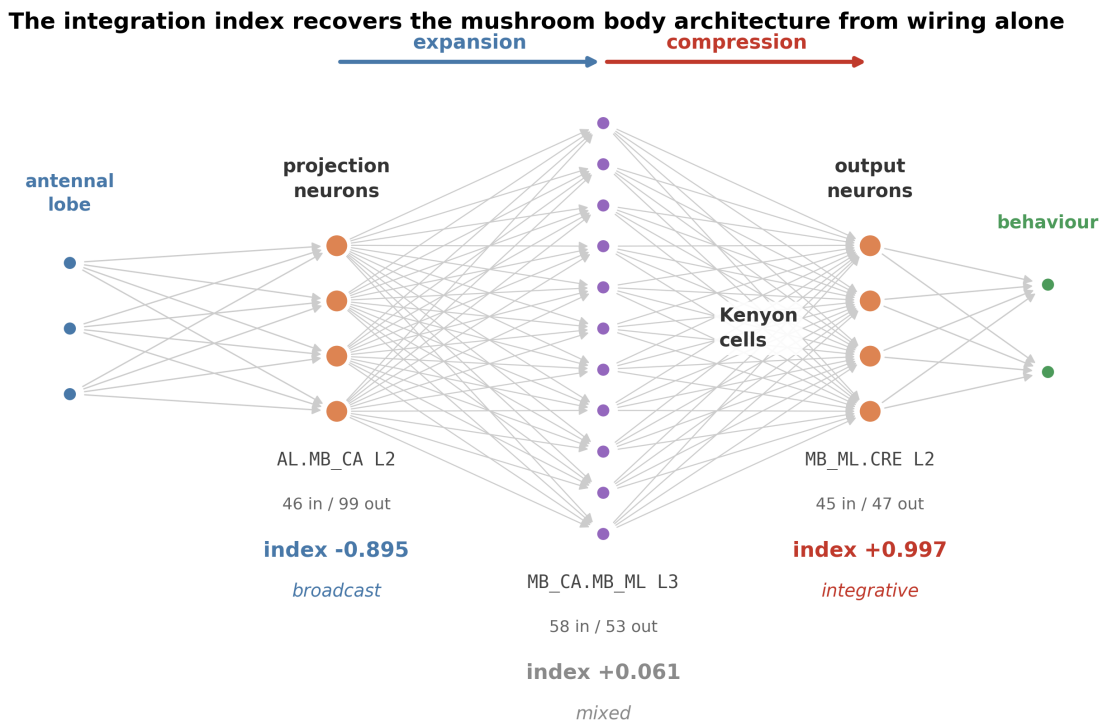


Figure 28: The olfactory pathway of the mushroom body, drawn so that the width of each stage reflects the operation the index attributes to it. Odour information arrives from the antennal lobe, is distributed by a small population of projection neurons onto a much larger one of Kenyon cells, and is then read out by a small population of output neurons. The integration index beneath each stage is computed from connectivity alone, with no anatomical information supplied: it comes out strongly negative where the circuit expands, near zero where it is maintained, and at the extreme of positive where it compresses. The values are those of Table 13.

The prediction is met, and in order, as Figure 28 shows. The two entries at the top of the table, the projection neurons and the parallel pathway into the lateral horn, come out strongly negative at -0.895 and -0.712 : they are broadcasts, distributing a compact odour representation outwards. The Kenyon cells sit almost exactly at the

boundary, +0.061, which is the mixed regime: they receive the expansion and pass it on without yet compressing it. The two output populations come out at +0.997 and +1.000, at the extreme of integration: they collapse the expanded representation back into a few channels. Read down the table, the index traces expansion, then maintenance, then compression, which is the textbook description of the circuit.

Three things follow, and they are worth separating because they are claims of different strength.

The first concerns the index rather than the biology. The ordering was recovered from connectivity alone, with no anatomical or physiological information supplied at any stage, and with the selection of rows fixed in advance. That is what makes it a test: the index could have come out flat, or in the wrong order, and it did not. This is the strongest of the three statements and it is a validation of the method.

The second concerns what the index adds to the rest of the thesis. The symmetric search of Section 4.4 identifies `MB_CA.MB_ML` and the mushroom body waists as high-count structures, but it describes them all the same way, since with $k_{\text{in}} = k_{\text{out}} = 2$ every pattern is symmetric by construction. The sweep separates them: the projection neurons and the output neurons are both bottlenecks, and both carry a great deal of mass, and yet they are performing opposite operations. The direction of the asymmetry is information the fixed-shape search discards, and recovering it costs ninety-four seconds.

The third is a caution about scope. The mushroom body is the case where the answer was known, which is precisely why it was used, and a method validated on one circuit has not thereby been validated on all of them. The remaining seven hundred metanodes carry index values too, and nothing here establishes that those values are equally meaningful. What the mushroom body licenses is the weaker and still useful claim that the index is capable of recovering a real computational organisation when one is present, which makes it worth applying elsewhere and worth checking against physiology wherever such a check is available.

4.11 Within-Area Characterisation

The analyses so far have studied interactions *between* anatomical areas. A complementary question is what the motif composition looks like *within* a single area, and whether areas differ from one another in that respect.

An area here is a neuropil, and a neuron is taken to belong to every neuropil named in its group assignment, so that a neuron labelled `AL.MB_CA` counts towards both the antennal lobe and the mushroom body calyx. Areas therefore overlap, which is anatomically correct: a projection neuron genuinely does arborise in both

structures. Of the forty-four neuropils in the dataset, thirty-five contain at least two hundred neurons and are analysed here.

Counting within a single area raises a difficulty that does not arise between areas. Reciprocal connections within one neuropil are frequent, so the sets of neurons available as inputs and as outputs overlap heavily, and the requirement that the five participating neurons be distinct can no longer be handled by simply choosing disjoint groups. The count is therefore computed in closed form with an inclusion–exclusion correction that subtracts the configurations in which neurons coincide. Explicit enumeration would be quite infeasible, requiring on the order of 10^{10} combinations in the medulla alone, whereas the closed form processes the entire connectome in 4.5 seconds. The formula was verified against exhaustive enumeration on three hundred random graphs, with exact agreement on all sixteen signatures.

4.11.1 What Survives the Null, and What the Null Removes

Across five hundred and sixty tests, being thirty-five areas by sixteen signatures, one hundred and one are significant at $q < 0.05$ after Benjamini–Hochberg correction. The comparison worth reporting, however, is not that count but what changes between the naive analysis and the controlled one.

Table 14: Enrichment before and after applying the within-area null model. The naive comparison against the global profile produces large values that the null removes entirely.

Area	Naive enrichment	Versus null	q
EB	12.6×	1.00×	1.00
MB_CA	—	0.76×	1.00
NO	13.5×	1.99×	0.028
FB	7.3×	1.84×	0.028

The ellipsoid body falls from an apparent twelve-fold enrichment to exactly one-fold: the entire effect was a consequence of how that area’s neurons are distributed across levels, and the null neutralises it completely. The noduli and the fan-shaped body retain a smaller but genuine enrichment of roughly twice the null expectation. The null model therefore does both of the jobs asked of it, removing false positives while preserving real signals, which is the property that makes the surviving results worth reporting.

Three findings survive the control. The antennal lobe shows 31% of its motifs in the LAT->LAT class, with its neurons spread across several levels rather than

concentrated in one, and it is the only genuinely informative lateral case in the dataset. It corresponds to the network of local interneurons that performs lateral inhibition between olfactory glomeruli, the spherical compartments of the antennal lobe in which the receptors for a single odorant type converge, which is a well-established circuit recovered here independently and not explicable by level concentration. The optic lobe shows the re-entrant medulla pattern at 57.3% and the lobula plate relay at 40.9%, consistent with Section 4.6.1 and reached by an entirely independent route. The associative areas, by contrast, show no comparable structure, which is itself informative: the lateral organisation observed elsewhere is specific to particular circuits rather than a generic property of dense tissue.

4.12 Computational Performance

Table 15 reports the effect of the sparse reformulation on the two most expensive waists of the search.

Table 15: Effect of the sparse reformulation on the most expensive bottlenecks.

Bottleneck	Before	After	Speed-up
ME_L2 ($ B = 24,835$)	3172 s	14.5 s	219×
ME_L3 ($ B = 10,709$)	355 s	1.4 s	254×

The single waist ME_L2 previously accounted for roughly half of the total runtime on the shallow window, which is what makes a change of this magnitude on one bottleneck worth the effort. With the sparse engine in place, no waist falls back to the reference loop on any of the deeper windows. End to end, the shallow window went from one hour and forty-four minutes to fourteen minutes and thirty-nine seconds.

Output distillation, described in Section 3.5.3, reduced the four windows from over 100 GB to 68 MB, a reduction of up to 18,847-fold on the largest window, while producing downstream results that are identical to those obtained from the full output. The distillation itself costs approximately thirty per cent of the search time, since the aggregates must be maintained as the rows are produced; this is the price of not writing the intermediate at all, and it is worth paying only because the search has become fast enough that regenerating the rows costs less than storing them.

5 Conclusions

5.1 Summary of Contributions

This thesis analysed the bow-tie architecture of the adult *Drosophila* connectome at two complementary scales, and its contributions fall into four groups.

The theoretical contribution is the characterisation of when motif counts compose, stated as Proposition 3.1. The boundary between linear counting and the $\Sigma\Pi$ formulation is the presence of branching, and cyclic patterns require quadratic memory and are therefore out of reach at connectome scale. The result is verifiable by exhaustive enumeration, was so verified on all thirty-one patterns up to four nodes, and applies beyond this particular connectome to any counting problem of the same shape. The accompanying catalogue also places the method honestly: already at four nodes, 87% of connected patterns fall outside what it can handle.

The computational contribution is to have made exhaustive counting tractable. Reformulating the inner loop as a matrix product delegated to optimised libraries, and then switching to a sparse product on the large waists where the dense form fails, produced a factor of 219 on the worst bottleneck and reduced the shallow window from one hour and forty-four minutes to under fifteen. Recognising that the row-level output is an intermediate rather than a result reduced storage from over 100 GB to 68 MB with no loss to any analysis.

The empirical contributions concern the architecture itself. The hourglass property is present, with a score of 0.498 under structural routing, but it depends on the routing scheme and is markedly weaker than in *C. elegans*, plausibly because the fly maintains several parallel modality-specific bottlenecks rather than one shared one. The deep brain is laterally dominated, with purely lateral branches carrying 82.6% of the motif mass against 38.0% in the shallow brain. The deepest level of the hierarchy is the most internally dense and yet marginal as a waist, a combination that points to recurrent rather than convergent organisation. And the two scales of analysis agree on the mid-hierarchy olfactory waists while diverging interpretably on input groups.

The methodological contribution is the way an initially broad hypothesis was narrowed to the claim the data actually support. An apparent neurochemical difference between backward and forward waists held on the shallow window, did not replicate on the others, and was traced to the regions the two classes of motif occupy; what remains is a specific and defensible statement about the medulla, and an open question about the central brain that needs a targeted design to answer. The related finding that the ranking between signatures inverts between null models

makes the same point from a different direction: single-null analyses of this kind can be confidently wrong.

5.2 Limitations

The hierarchical decomposition on which everything else rests is inherited from earlier work rather than derived here, and it is only partly tested. The functional gradient of Section 4.1 supports the spectral embedding that produces the underlying axis, but it cannot validate the four thresholds placed on that axis, because those thresholds were themselves chosen to separate the super-classes whose gradient is then observed. Section 4.2 establishes that the levels are at least as compatible with the anatomical grouping used here as with the one they were built for, which addresses a different concern. Where the boundaries fall therefore remains an untested inheritance, and any result conditioned on levels carries whatever assumptions went into producing them.

The hourglass score depends on the routing scheme, taking values between 0.247 and 0.498 depending on the scheme and the path-length bound. No single number should be quoted in isolation, and the comparison with *C. elegans* should be read with the same caution, since the two studies do not use identical routing conventions.

The coarse motif classification is close to vacuous, with 94.4% of patterns falling into the *mixed* bucket. The sixteen-signature decomposition addresses this, but even after it the residual `mix->mix` class retains 29.5% of patterns in the shallow window. Resolving that class further would require distinguishing branches by which specific levels they connect, at the cost of a considerably larger label space.

No null model isolates the hypothesis cleanly. The three used here preserve different constraint sets and none is fully satisfactory: model A preserves global degrees but destroys anatomy, model B preserves anatomy but destroys degrees, and model C preserves both while having a ratio dominated by sparsity. A null that simultaneously preserved within-block degrees and block densities would make the statistic essentially deterministic, since the motif count is very nearly a deterministic function of the degree sequences. The limitation is intrinsic to the quantity being tested rather than to the implementation, and Section 4.7.1 shows that it has real consequences for interpretation.

Realised compression remains sensitive to a collapsing waist. When the flow is carried by a single neuron the ratio explodes, with `L0_L3` reaching 6316 at an effective waist of 1.3 and `ME_L2` reaching 4259 at 1.8. Such structures are hubs rather than waists and are excluded here by a threshold on the effective waist size, set to five throughout, but that threshold is a declared choice and results should be

checked for robustness against it. Reporting the pair consisting of the effective size and the periphery, instead of their ratio, would avoid the division entirely and is worth considering.

The pattern catalogue stops at four nodes, since exhaustive enumeration on five would require 2^{20} edge masks over 120 permutations. The classification of Proposition 3.1 does not depend on pattern size, so the qualitative conclusion carries, but the proportion of non-factorisable patterns beyond four nodes is not measured here. Relatedly, that proposition concerns homomorphisms, whereas the requirement that participating neurons be distinct is a separate inclusion–exclusion layer; the two should be kept conceptually apart.

The sweep over k_{in} and k_{out} covers only the pure forms, because the symmetric-polynomial shortcut it relies on admits an exact disjointness correction only for those. This is a limitation of the sweep and not of the search engine, which counts any configuration exactly; what limits the engine at larger k_{in} and k_{out} is the combinatorial growth of the number of group combinations, not correctness.

The distillation fixes the selection criteria at search time, since the retained top- N selections are chosen by occurrence count. An analysis requiring a different criterion, such as selection by realised compression, would require re-running the search rather than merely re-reading the stored output.

Finally, and most fundamentally, all the results here are structural. Connectivity constrains computation but does not determine it, and no claim about neural dynamics follows directly from these counts. What a structural analysis can establish is which circuits exist and in what abundance; what they do remains a question for physiology.

5.3 Future Directions

The most immediate extension is to carry the remaining analyses through to the deep windows. Realised compression and statistical validation were conducted in depth only on the shallow window, which is the least rich of the four with thirty-one million motifs against four hundred and thirty-four million, and there is no methodological obstacle to repeating them now that the search is fast enough.

The open question left by Section 4.9 calls for a targeted design rather than more data. Testing whether backward waists are more inhibitory than forward ones in the central brain requires a set of waists chosen in advance to carry both directions in comparable numbers, since the present comparison is confounded by the regions the two classes of motif happen to occupy. An exhaustive search cannot assemble such a set, because it takes the waists the connectome provides; a targeted search over

waists pre-selected for balance could, and would settle whether the medulla result generalises or is specific to motion-detection circuitry.

The behaviour of level 5 deserves separate investigation. It is the most internally dense level of the connectome and yet marginal as a waist, and characterising the connectivity that produces this combination, presumably recurrent and lateral rather than convergent, is a well-posed structural question with a plausible functional interpretation attached to it.

The framework already supports arbitrary values of k_{in} and k_{out} , so a systematic exploration of asymmetric configurations would connect the bow-tie analysis to the integrative and broadcast distinction of Section 4.10 in a more thorough way than the present sweep allows.

Finally, both the counting framework and the hourglass measurement are species-agnostic. Applying them to other fully mapped connectomes as those become available would establish whether the architectural principles found here are general or particular to the fly, which is the question that motivates structural connectomics in the first place.

References

- [1] D. S. Bassett and O. Sporns. Network neuroscience. *Nature Neuroscience*, 20(3):353–364, 2017.
- [2] Y. Benjamini and Y. Hochberg. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1):289–300, 1995.
- [3] E. Bullmore and O. Sporns. Complex brain networks: graph theoretical analysis of structural and functional systems. *Nature Reviews Neuroscience*, 10(3):186–198, 2009.
- [4] L. P. Cordella, P. Foggia, C. Sansone, and M. Vento. A (sub)graph isomorphism algorithm for matching large graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 26(10):1367–1372, 2004.
- [5] E. Corradini, F. Parlapiano, A. Ronci, G. Terracina, and D. Ursino. A complex network-based approach to detect and investigate connectome motifs in the larval *Drosophila*. *Computers in Biology and Medicine*, 192, 2025.

- [6] D. Curcio. *Decoding the Drosophila Brain: A Parametric Framework for Hierarchical Motif Discovery in the Adult Connectome*. Master’s thesis, University of Applied Sciences Upper Austria and University of Calabria, 2026.
- [7] K. Ito et al. A systematic nomenclature for the insect brain. *Neuron*, 81(4):755–765, 2014.
- [8] A. Lin et al. Network statistics of the whole-brain connectome of *Drosophila*. *Nature*, 634(8032):153–165, 2024.
- [9] S. Maslov and K. Sneppen. Specificity and stability in topology of protein networks. *Science*, 296(5569):910–913, 2002.
- [10] R. Milo, S. Shen-Orr, S. Itzkovitz, N. Kashtan, D. Chklovskii, and U. Alon. Network motifs: simple building blocks of complex networks. *Science*, 298(5594):824–827, 2002.
- [11] K. M. Sabrin, Y. Wei, M. P. van den Heuvel, and C. Dovrolis. The hourglass organization of the *Caenorhabditis elegans* connectome. *PLOS Computational Biology*, 16(2):e1007526, 2020.
- [12] M. Tricoli. *Structural Modeling of the Drosophila melanogaster Connectome: Motif-Based and Hierarchical Analysis*. Master’s thesis, University of Calabria, 2025.
- [13] J. R. Ullmann. An algorithm for subgraph isomorphism. *Journal of the ACM*, 23(1):31–42, 1976.
- [14] M. Winding et al. The connectome of an insect brain. *Science*, 379(6636):eadd9330, 2023.